Laboratory 4
Bank Office model

AnyLogic provides the Process Modeling Library, a discrete-event simulation library containing objects you can use to rapidly simulate complex discrete-events systems like:

- Manufacturing processes with detailed shop floor layout
- Simple and complex service systems (e.g. banks, airports, etc.)
- Business processes with activity based costing
- Logistics and supply chain models

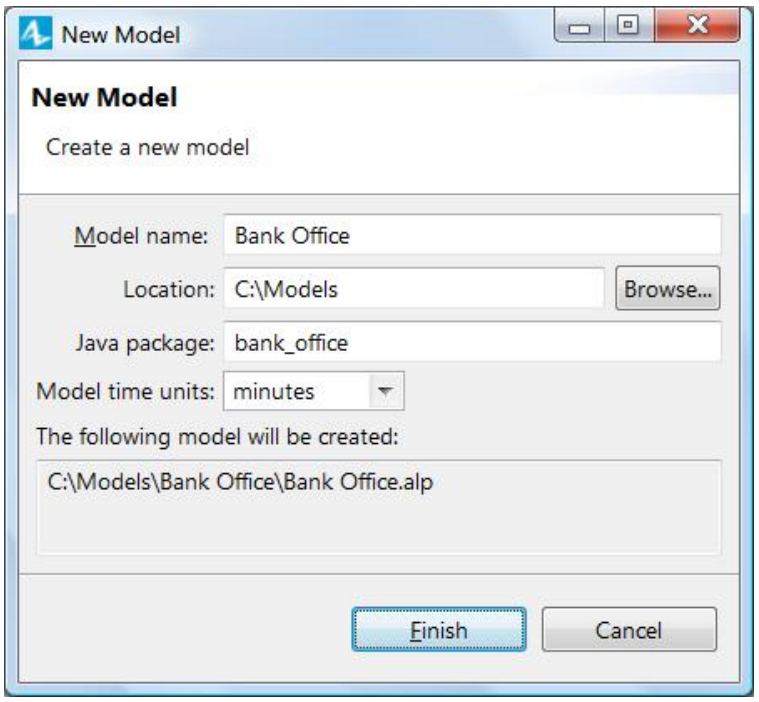
The Process Modeling Library allows you to create flexible models, collect basic and advanced statistics, and effectively visualize the process you are modeling to validate and present your model.

In this laboratory we will create a simple service system of a bank department, consisting of an automatic teller machine and teller lines. ATM provides people with a quick self-service for cash. More complex transactions, e.g. paying bills, are completed by tellers, allowing customers more time without inconveniencing those customers looking for quick cash.

Phase 1. Creating a Simple Model

First, we will create the simplest queueing model simulating how customers are serviced at the ATM.

1. Create a new model by selecting **File** > **New** > **Model** from AnyLogic's main menu. The **New Model** dialog box is displayed.
2. Specify the name of the model. Type *Bank Office* in the **Model name** edit box.



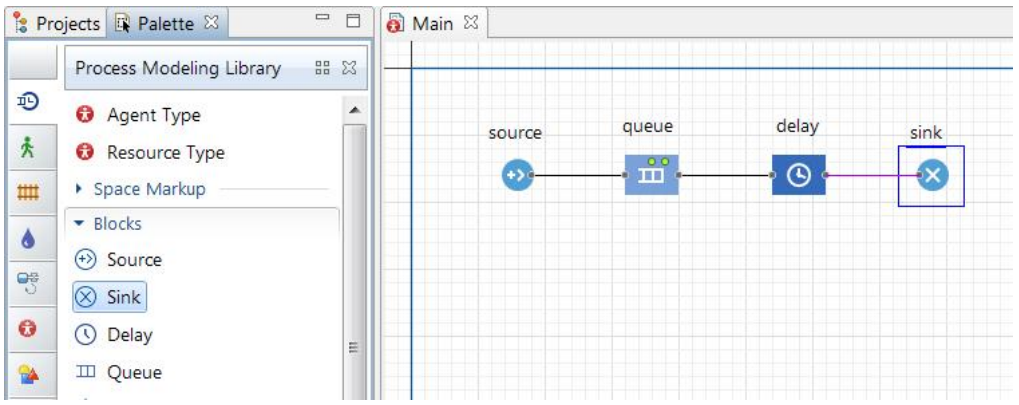
3. Specify the location where you want to store your model files. Browse for the existing folder using the **Browse** button, or type the name of the folder you want to create in the **Location** edit box.

4. Select *minutes* as the **Model time units**.

5. Click **Finish** to complete the process.

6. After you create the model, AnyLogic will switch to the **Palette** view and display the **Process Modeling Library** palette.

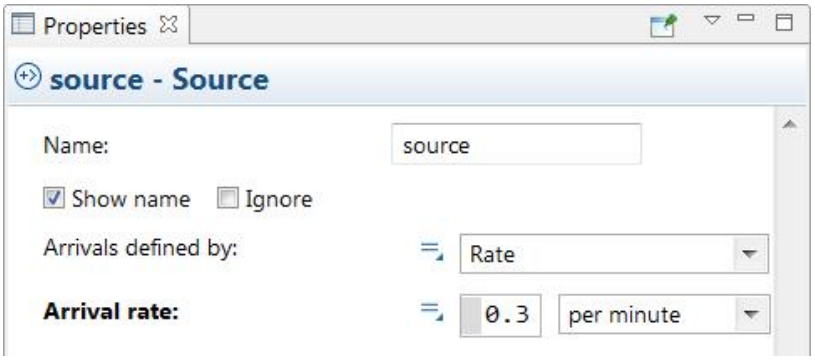
7. Add **Process Modeling Library** blocks on the diagram and connect them as shown in the figure below. To add a flowchart block on the diagram, drag the required element from the palette into the graphical editor.



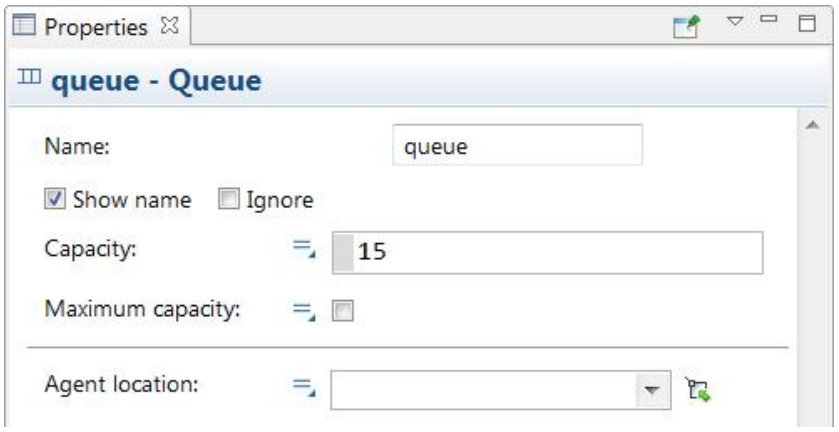
8. While dragging the blocks, and placing them close to each other, you may see the lines connecting the blocks appear. Please note that these connectors should connect only the ports lying on right and left borders of the block icons.

9. To modify properties of some model element, first select it by clicking on it in the graphical editor or in the **Projects** view. This opens the properties of this element in the **Properties** view.

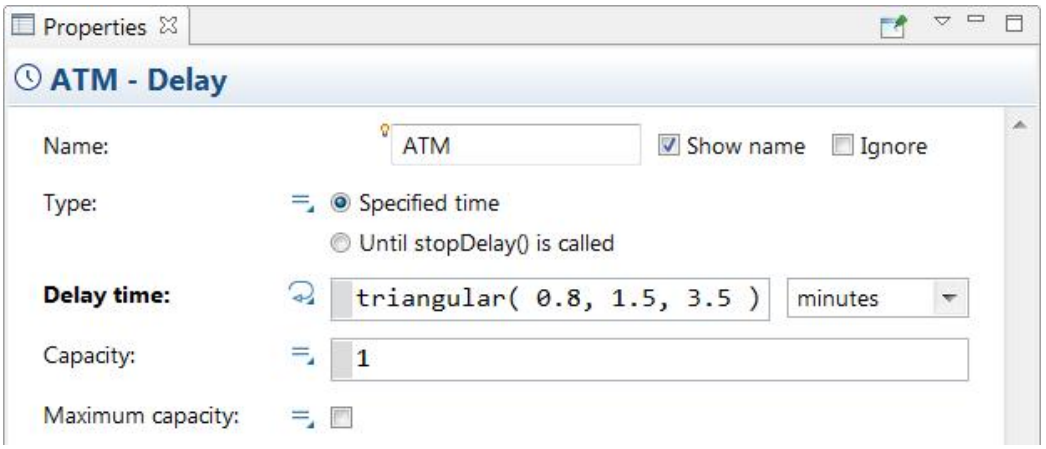
10. Select *source* block. In the **Properties** view, specify how often customers arrive. Type *0.3* and select *per minute* for **Arrival rate**.



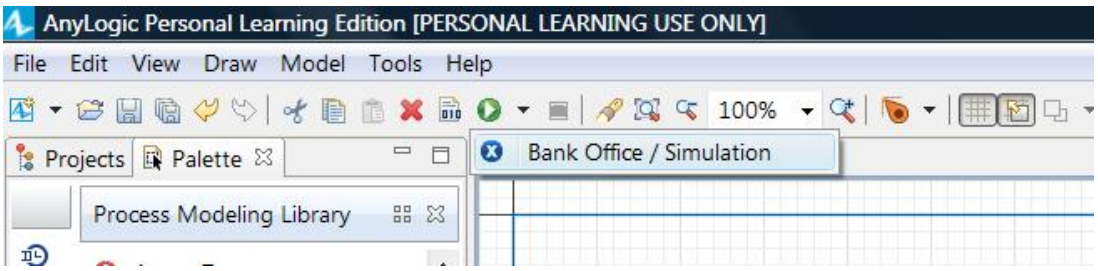
11. Modify the properties of the *queue*. Set queue capacity to *15* agents. At most *15* customers will wait in a queue.

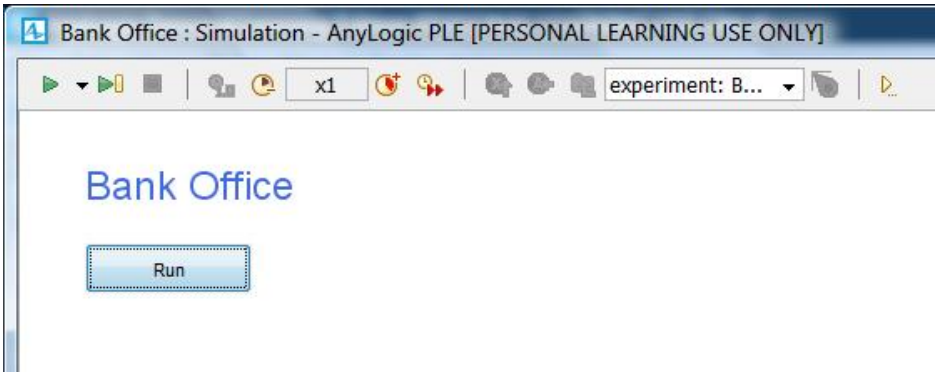


12. Modify the properties of the *delay*. Name the object *ATM*. Specify the processing time. Assume that processing time is triangularly distributed with mean value of *1.5*, min of *0.8* and max value of *3.5 minutes*.

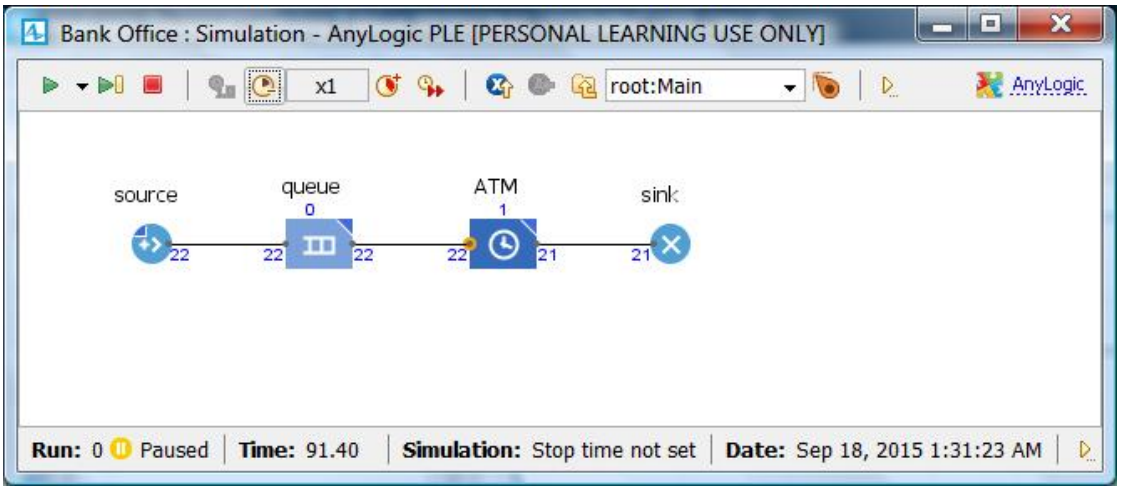


13. Click the **Run** toolbar button and choose the experiment you want to run from the drop-down list. Your simulation experiment is called **Bank Office/Simulation**.



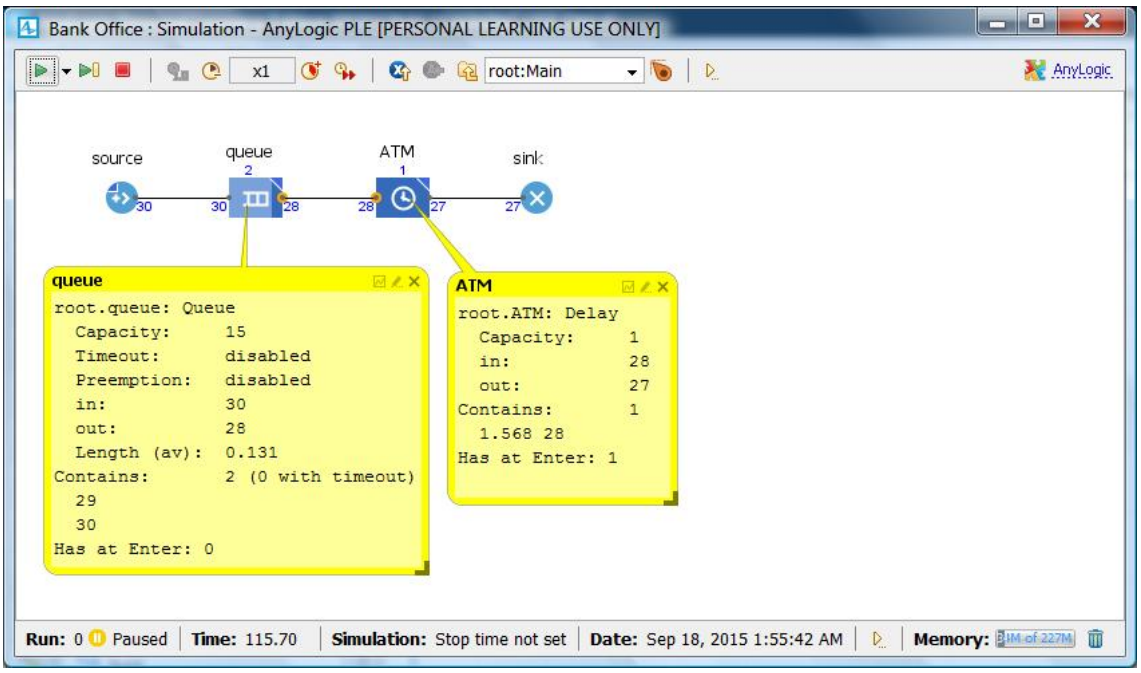


Click the button to **Run** and observe the process dynamics. You will see the animated flowchart. Each model created with **Process Modeling Library** instantly has animated flowchart where you can see detailed current object status, for example queue size, number of agents left and so on – completely in graphics!



If needed, adjust the execution speed to your needs using **Slow down** and **Speed up** toolbar buttons.

You can inspect flowchart objects to get the detailed information on their current state. Click on the object to open its inspect window. Inspect window show statistics on the object, e.g. *queue* object's inspect shows the queue capacity, the number of agents passed through either port of the object and also whether the timeout option is enabled for this queue. **Contains:** string displays the number of agents currently being in the object along with IDs of these agents.



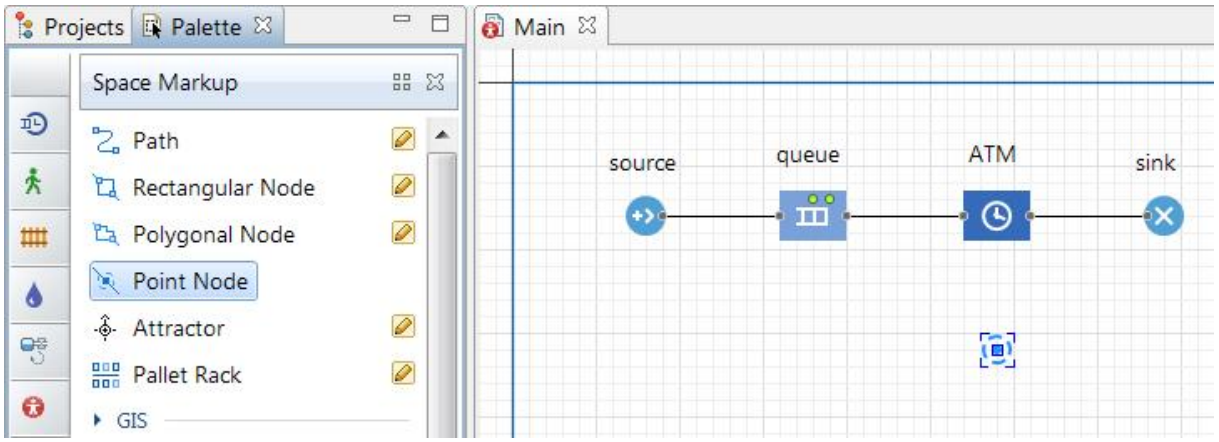
Phase 2. Creating Model Animation

Although the flowchart is animated, you may want to see the actual bank department layout animated. That is also possible! For each model you can create an animation to visually represent your model. You can create any animation you want. Now we will draw the layout consisting of the ATM and a queue. Then we will animate clients standing in the queue and using the ATM. We also want to visualize the current status of the ATM.

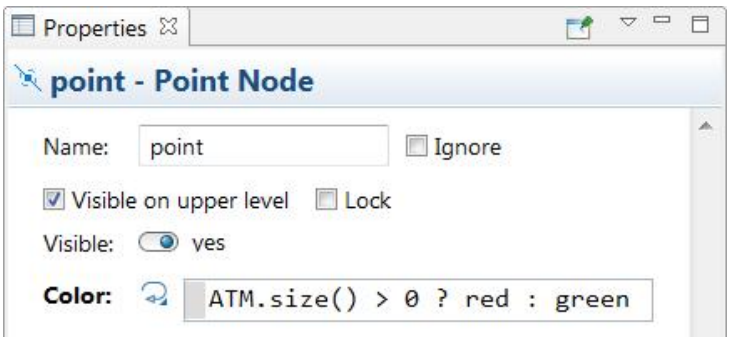
Now we will draw the layout of our bank. You draw the layout on the same diagram where you draw a flowchart.

1. Draw the ATM as a point node. First, open the **Space Markup** palette in the **Palette** view.

2. Drag the element **Point node** from the **Space Markup** palette into the graphical editor and place it under the flowchart.

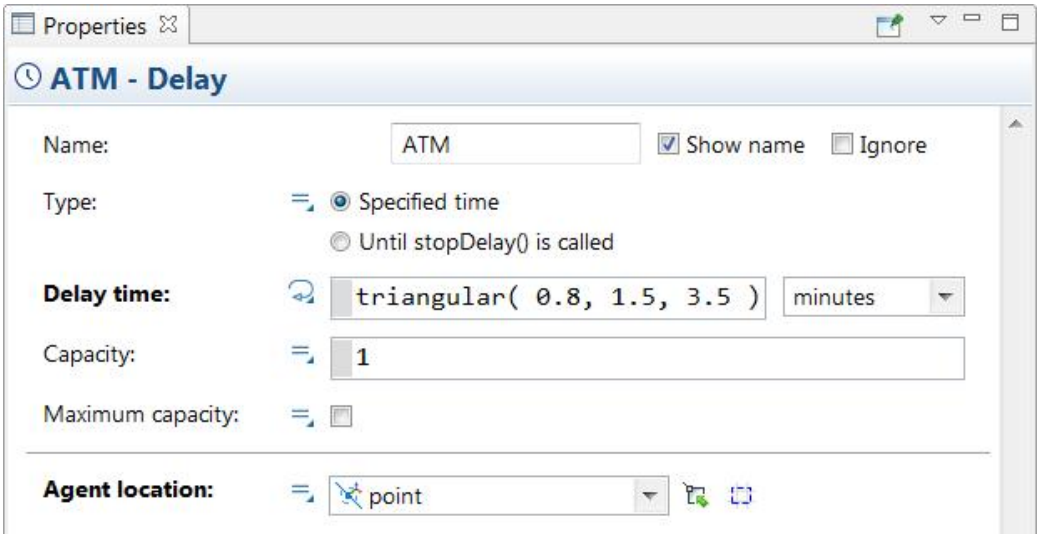


3. Select the point node in the graphical editor to open its **Properties** panel. Enter the run-time color expression for the shape in the **Color** property: *ATM.size() > 0 ? red : green.*



4. Click the *delay* block, called *ATM*, in the flowchart to open its **Properties** view.

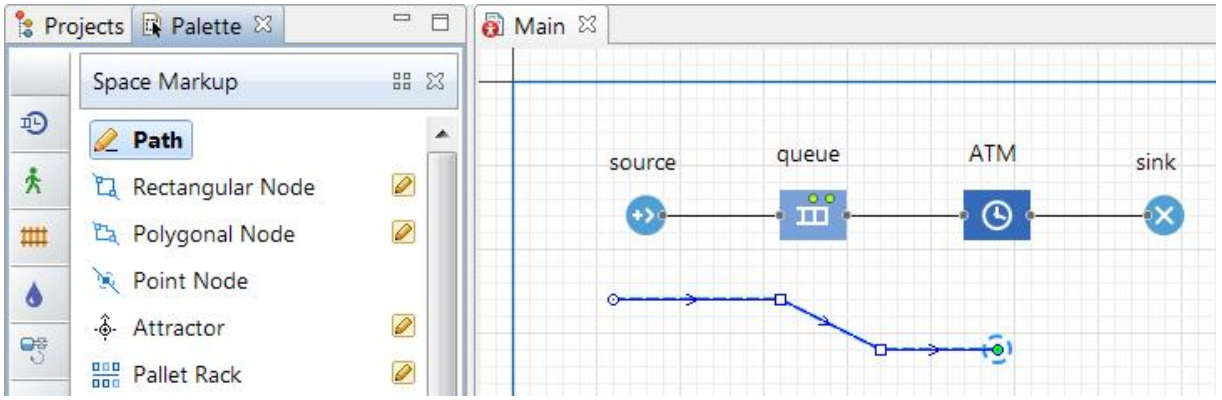
5. Select the *point* node we have drawn in the **Agent location** option. You can either click the down arrow and select the point node from the list of appropriate objects, or you can click the button, located on the right, to select this space markup shape from the graphical editor (the rest of the elements in the editor will be greyed out).



6. Draw the queue as a path. First, open the **Space Markup** palette in the **Palette** view.

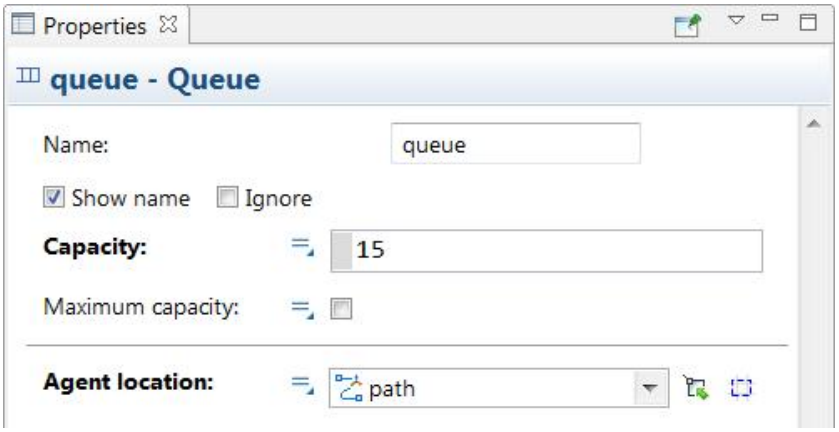
7. Double-click the element **Path** to switch to the drawing mode.

8. Click in the graphical editor to put the first point of the path. Do more clicks to add turning points. Finish drawing with a double-click.



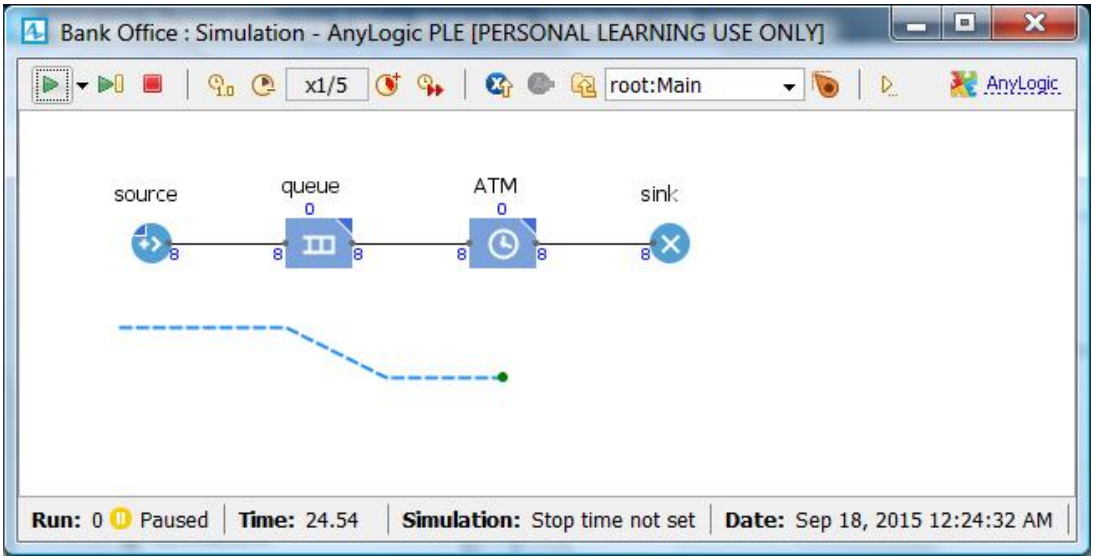
9. Click the *queue* block in the flowchart and go to its **Properties** view.

10. Select the *path* we have drawn in the **Agent location** option. You can either click the down arrow and select the path from the list of appropriate objects, or you can click the button, located on the right, to select this space markup shape from the graphical editor (the rest of the elements in the editor will be greyed out).



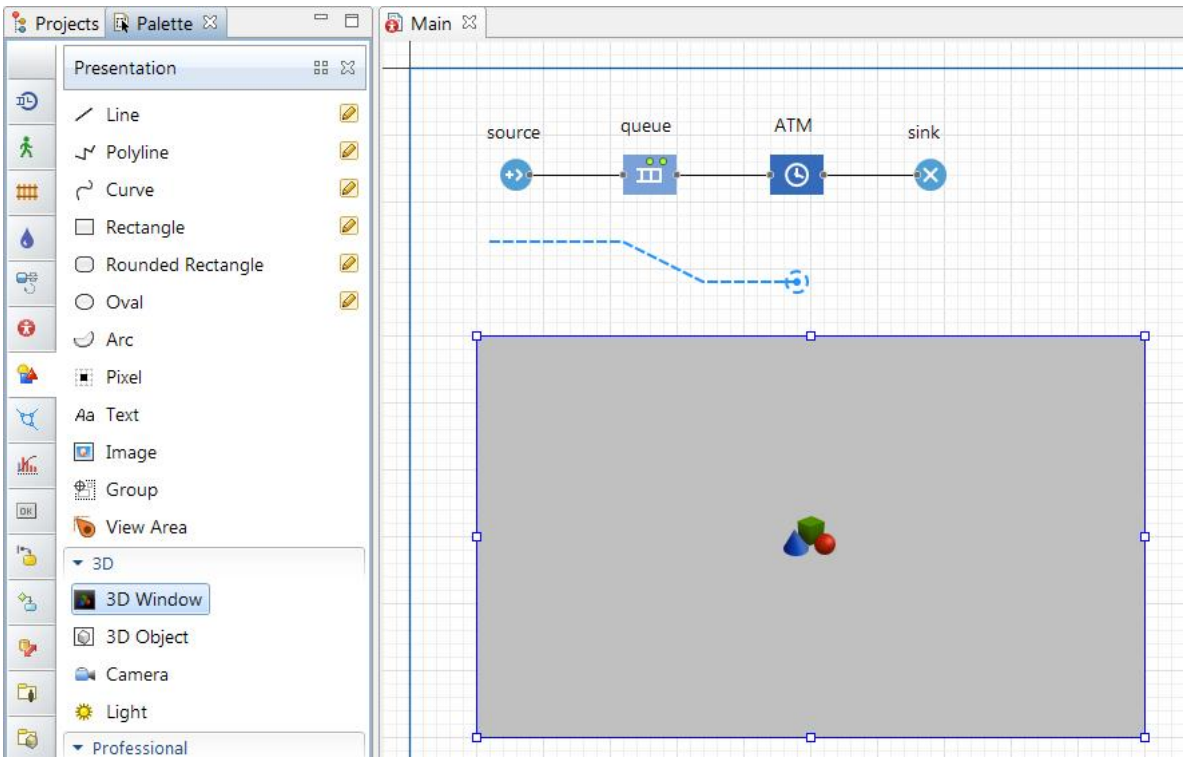
Now you can run the model and observe its behavior. If you want to speed up the simulation significantly, switch to virtual time mode by clicking the **Toggle real/virtual time** mode toolbar button. Switching to virtual time mode allows you to view simulation run at its maximum speed. Therefore, you can simulate a long period of time.

Note that when the ATM station is serving a customer, its animation shape becomes red, and when it is idle, it is green.



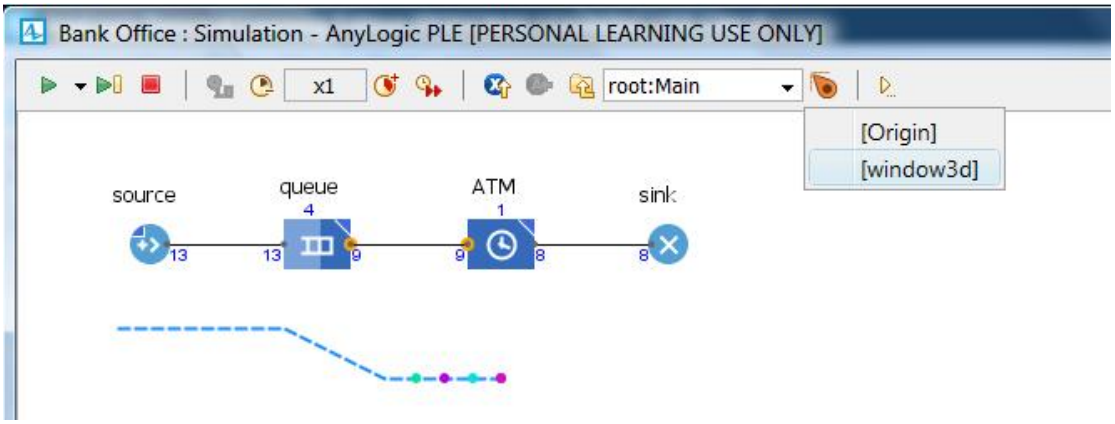
11. Drag the **3D Window** element from the **3D** section of the **Presentation** palette to the graphical editor.

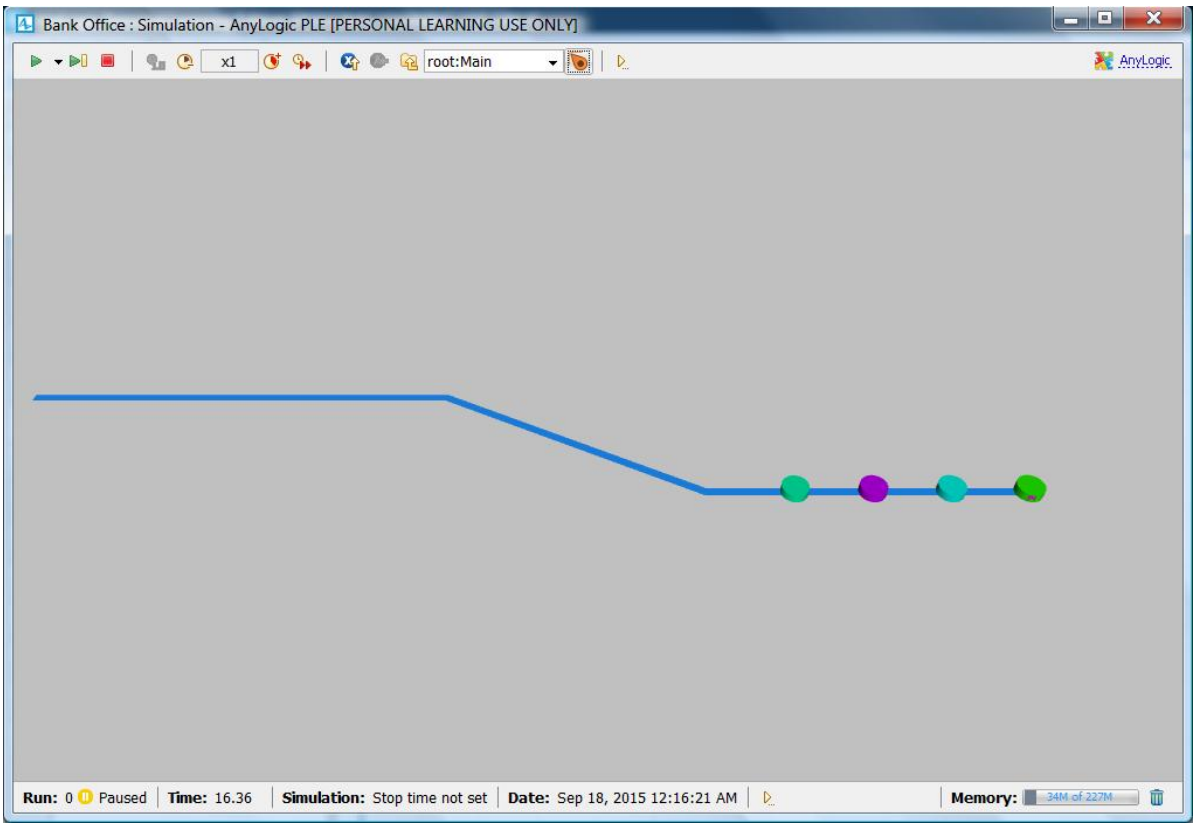
12. You will see the grey area appeared on the screen. Locate it where you want your 3D presentation to be shown at the model runtime:



Now you can run your model and observe simple 3D animation.

13. Click the toolbar button **Navigate to view area...** and select [**window3D**].

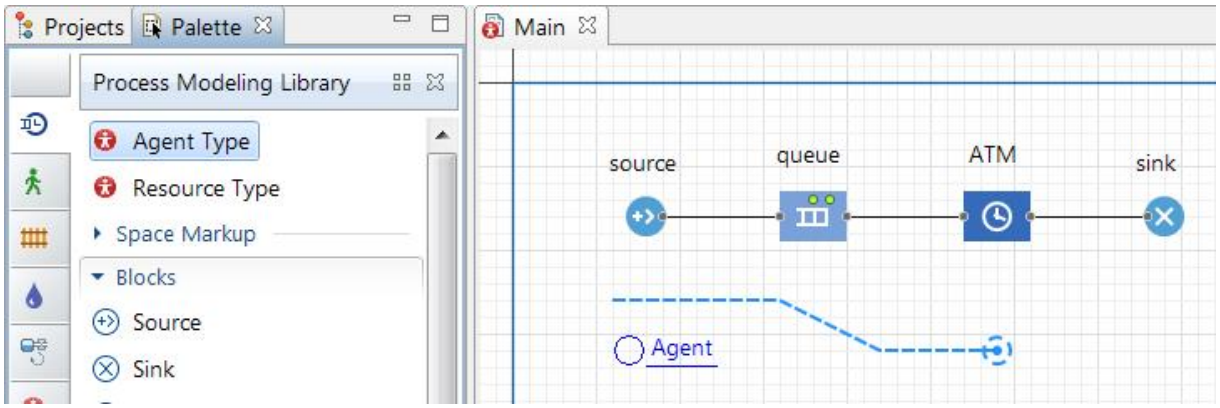




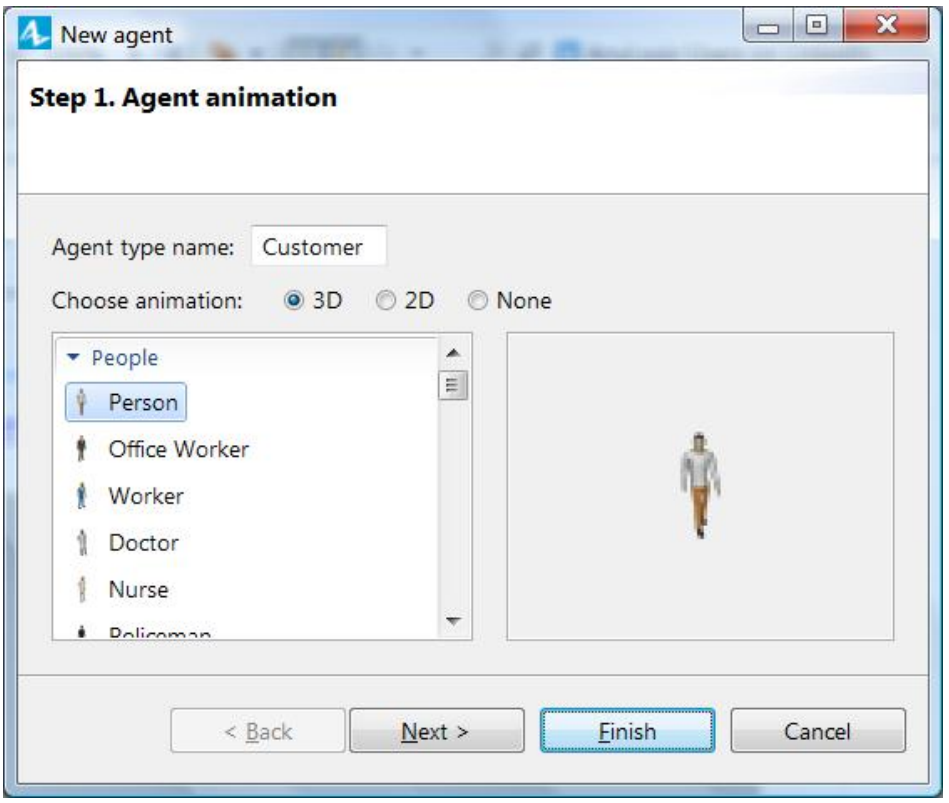
Now we want to add a customer figure. By default the customers are drawn as coloured dots and shown as colored cylinders in 3D animation. We want to create our custom type of client and animate it in 3D. We will create a new agent type for this purpose.

14. Open the **Process Modeling Library** in the **Palette** view.

15. Drag the element **Agent Type** into the graphical editor.



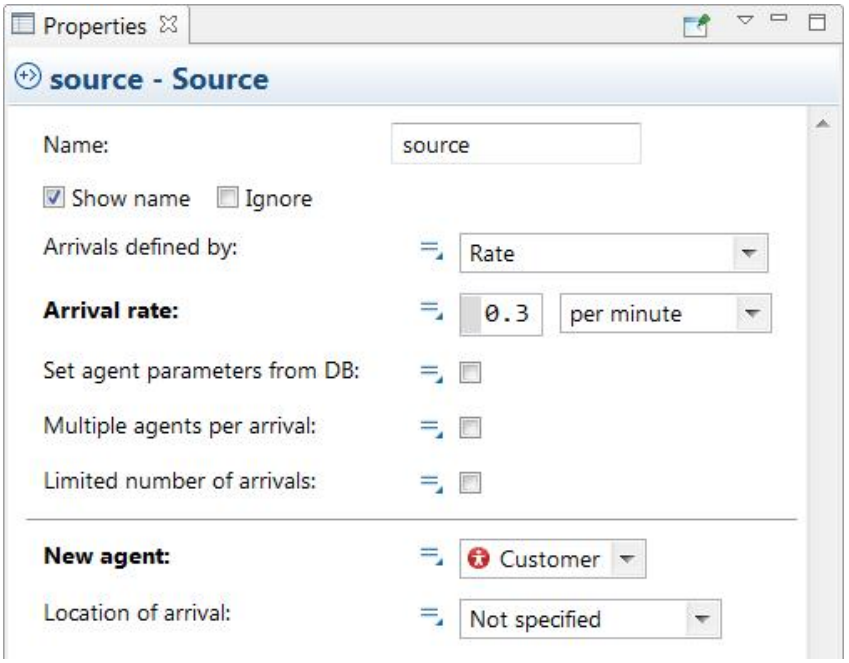
16. The **New agent** wizard will open on the **Agent animation** step. Enter *Customer* as the **Agent type name**, select **3D** as the animation type and select *Person* from the list of the 3D figures.



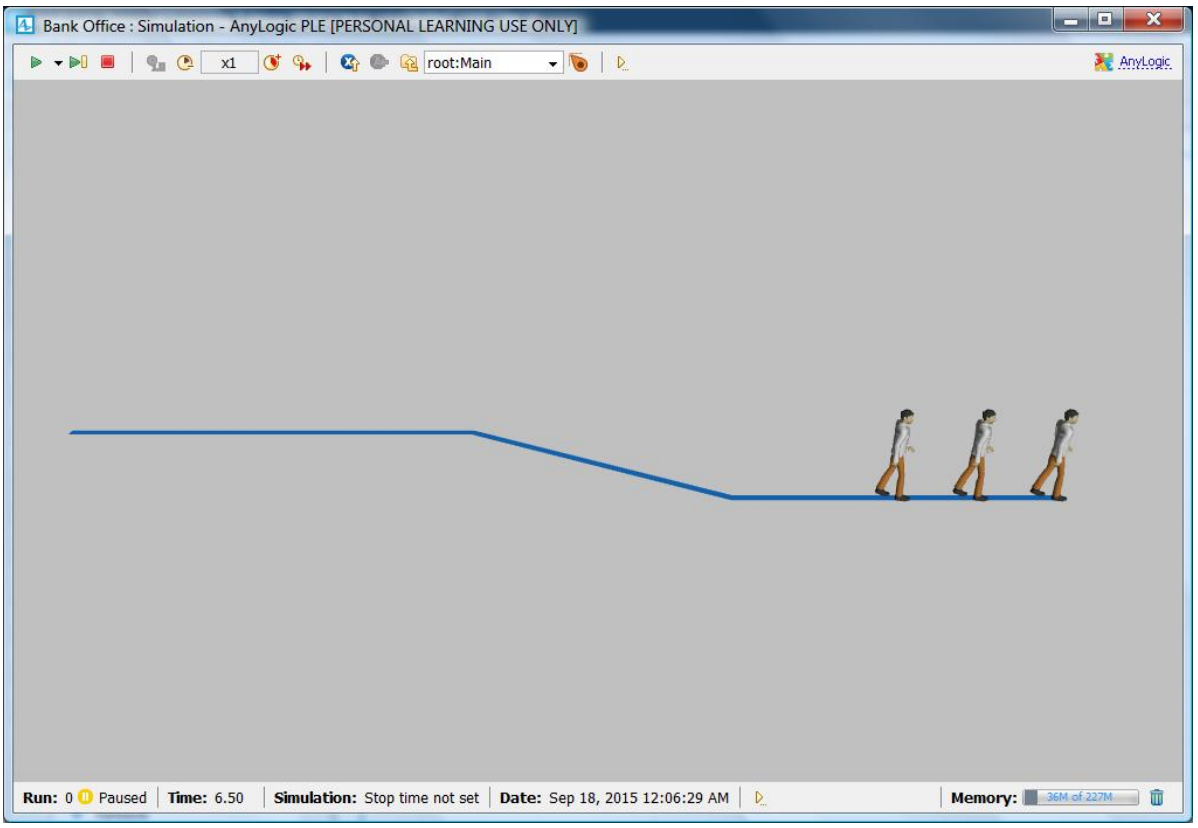
17. Click **Finish**. The new *Customer* diagram will open. You can find the *Person* 3D figure in the axis origin. Switch back to the *Main* diagram.

18. On the *Main* diagram, select the block *source* in the graphical editor.

19. Choose *Customer* in the **New agent** drop-down list.

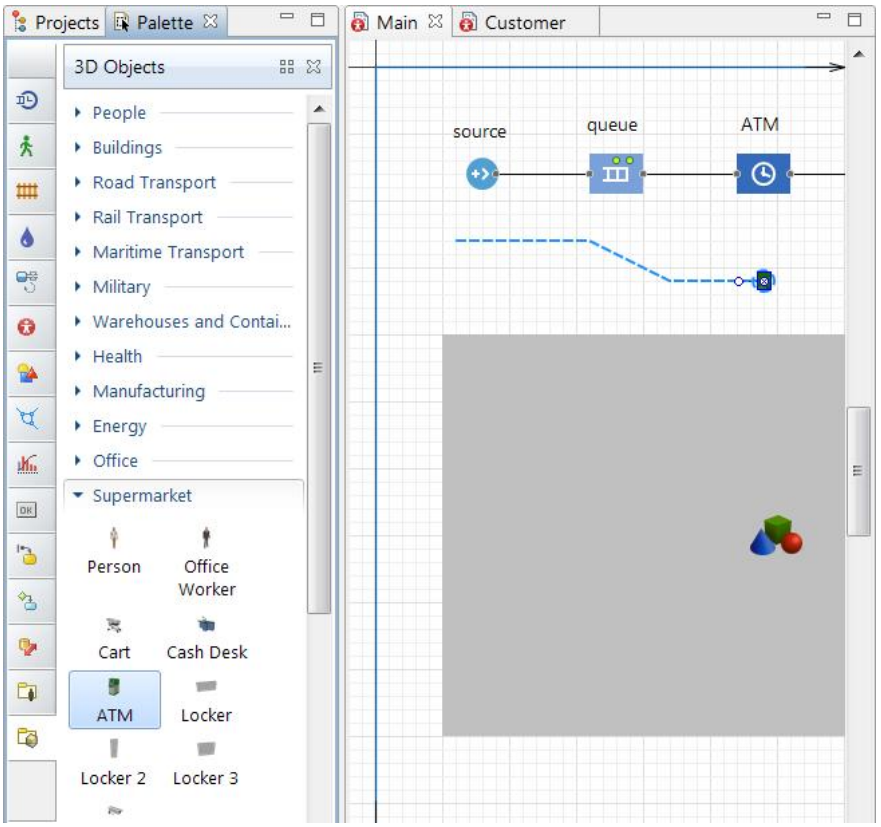


20. Run the model and switch to 3D view to see our customers moving in the queue.



21. Open the **3D Objects** palette in the **Palette** view.

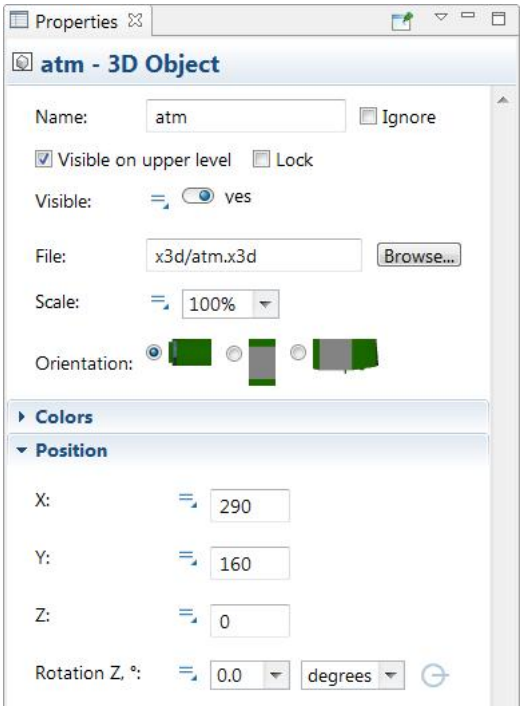
22. Drag the **ATM** 3D figure from the **Supermarket** section of this palette onto the point node shape in the graphical editor.



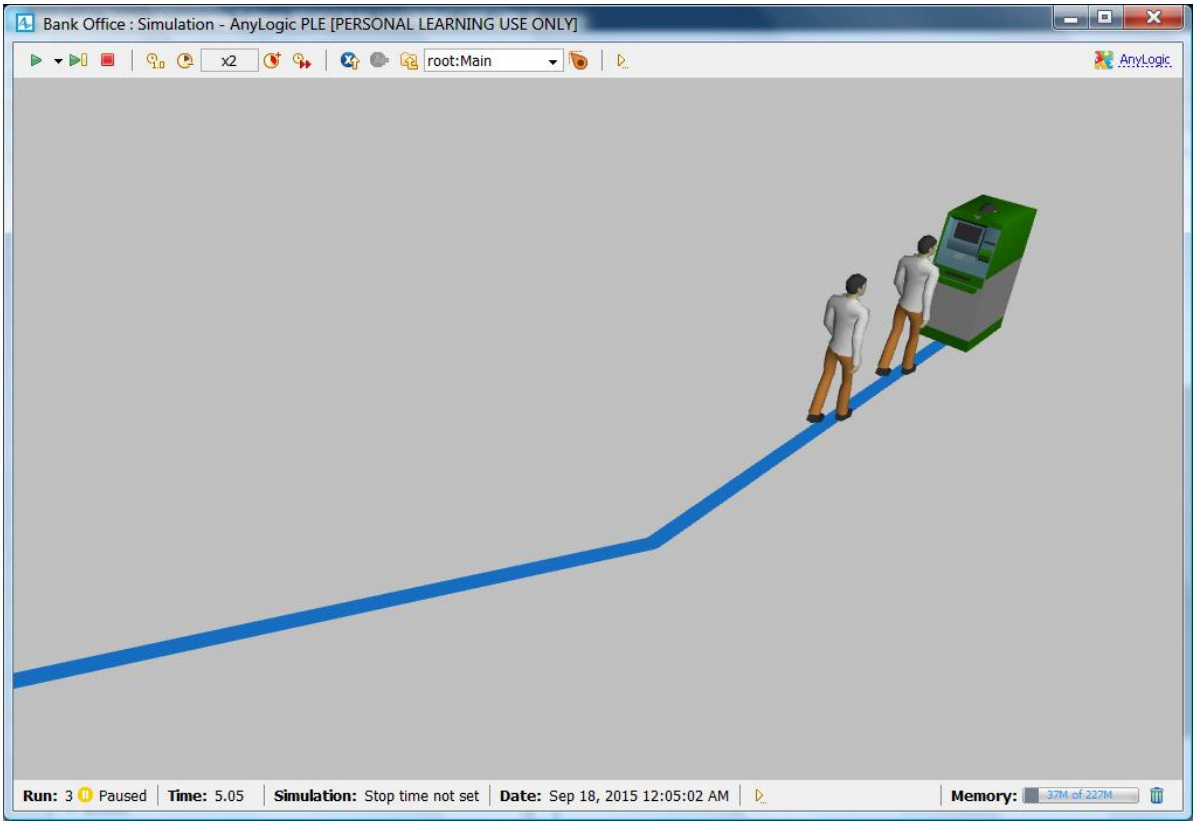
23. If you run the model now and check 3D animation in **window3D** mode, you will notice that our ATM does not face the customers' flow and we need to rotate it.

24. Select the *atm - 3D object* in the graphical editor and open the section **Position** in its properties view.

25. Choose *0 degrees* from the drop-down list of the **Rotation Z** option.



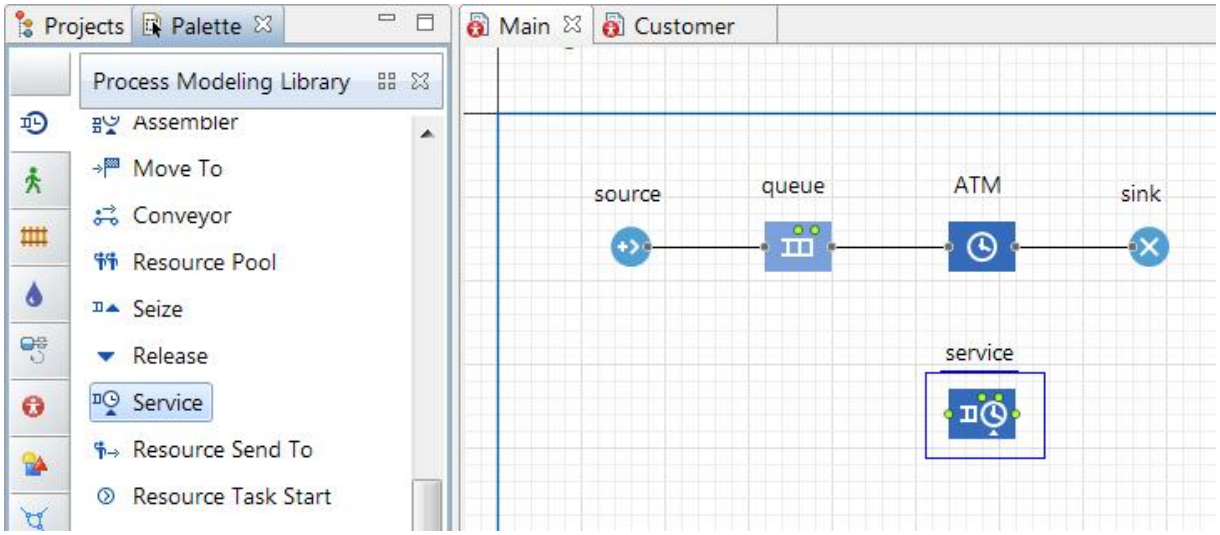
26. Run the model to double-check that the ATM is facing the customers now.



Phase 3. Adding Tellers

Now we will create another part of the system by adding tellers that are working at the bank. Now some clients will come to see tellers, some – to access the ATM. We can model tellers using delays in the same way as we modeled ATM. However, modeling tellers using resources is much more convenient. *Resource* is a special unit that can be possessed by an agent. Only one agent can possess a resource at a time; therefore agents compete for resources.

1. Open the **Process Modeling Library** in the **Palette** view and drag the **Service** block onto our *Main* diagram.

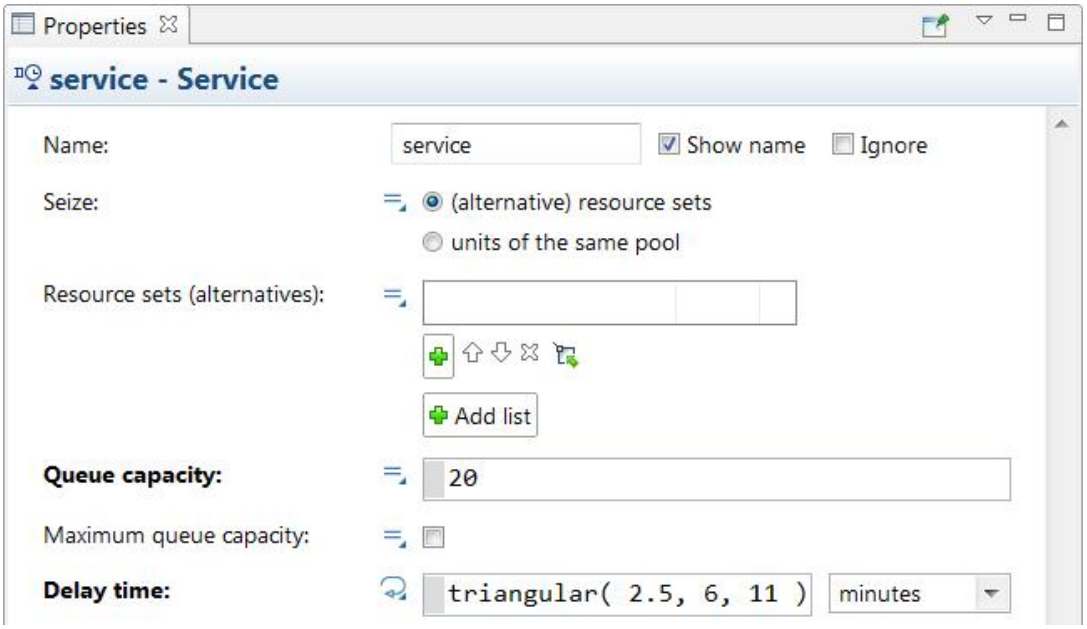


   Service seizes resource units for the agent, delays the agent, and releases the seized units.

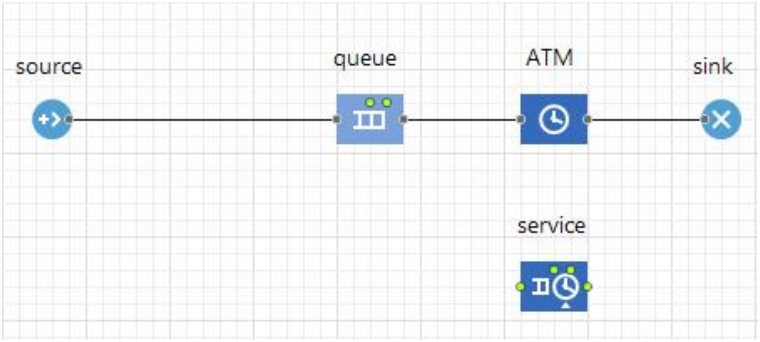
2. Go to the **Properties** view of the *service* block.

3. Modify the object properties:

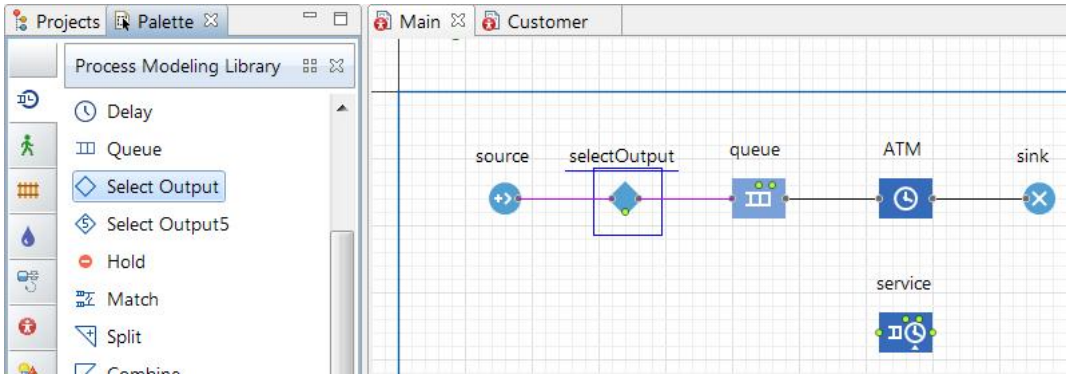
   - There is one queue for all tellers. Set up **Queue capacity** to be of *20* places.
   - We assume that service time is triangularly distributed with the min value of *2.5*, average value of *6*, and the max value of *11 minutes*. Set **Delay time**: *triangular(2.5, 6, 11) minutes.*



4. Move the objects *queue*, *ATM* and *sink* to the right to make space for one new object between *source* and *queue*.

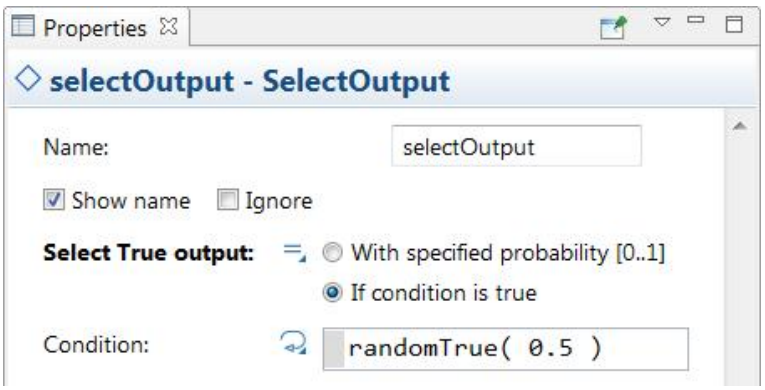


5. Open the **Process Modeling Library** in the **Palette** view and add the **SelectOutput** block in the resulting space. When you place the object on the connector, it will automatically get built in.

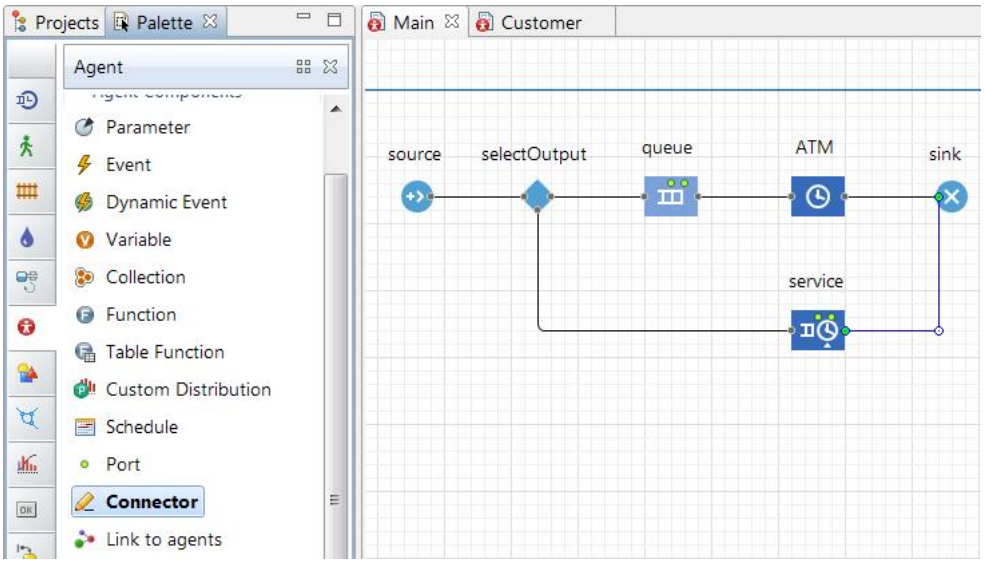


SelectOutput object is a decision making block. The agent arrived at the object is forwarded along one of two output ports depending on the user-defined condition.

6. Select *selectOutput* in the flowchart and go to its **Properties** view. Choose the option *If condition is true* for the **Select True Output** parameter. Make sure that **Condition** is *randomTrue(0.5)*. This agent routing condition defines that the number of customers competing for ATM and teller service will be approximately equal.



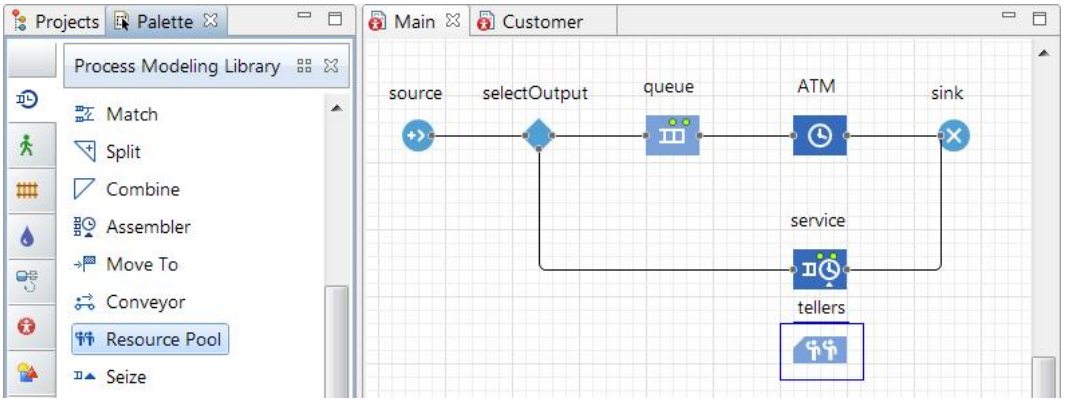
7. Connect *selectOutput* and *service* with other blocks as shown in the figure:



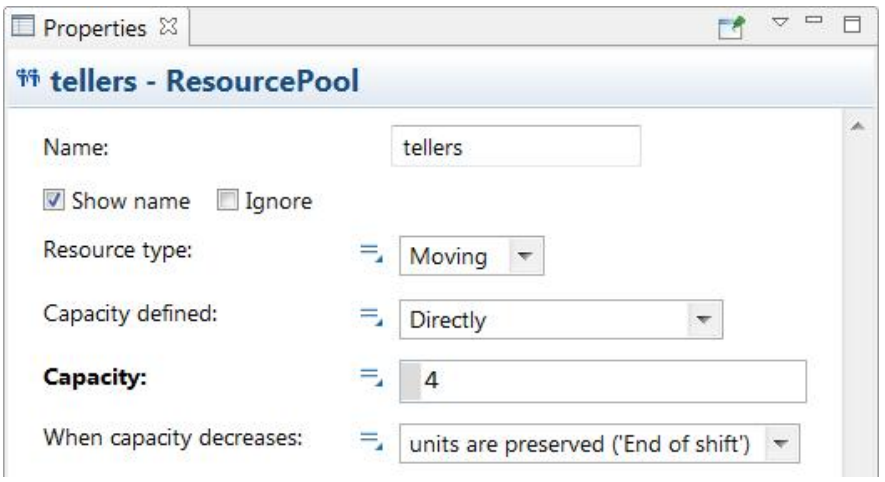
8. Open the **Process Modeling Library** in the **Palette** view and drag the **ResourcePool** block onto our *Main* diagram. **ResourcePool** object is storage for resource units.

9. Place it under *service* and go to its **Properties** view.

10. Name the object *tellers*.



11. Specify that this resource object has only four resource units, that means, define its **Capacity**: *4*.



12. **ResourcePool** object should be connected to resource seizing and releasing objects (**Service** in our case). So we need to modify the properties of the *service* object.

13. Select *service* in the flowchart to open its properties. Choose the option *units of the same pool* the parameter **Seize**. Then specify the resource pool we have created in the option **Resource Pool**. You can either click the down arrow to select the resource pool object from the drop-down list, or you can click the button, located on the right, to select the object in the graphical editor (all inappropiate objects will be greyed out).



14. Now since the model has changed, we need to alter the model animation as well.

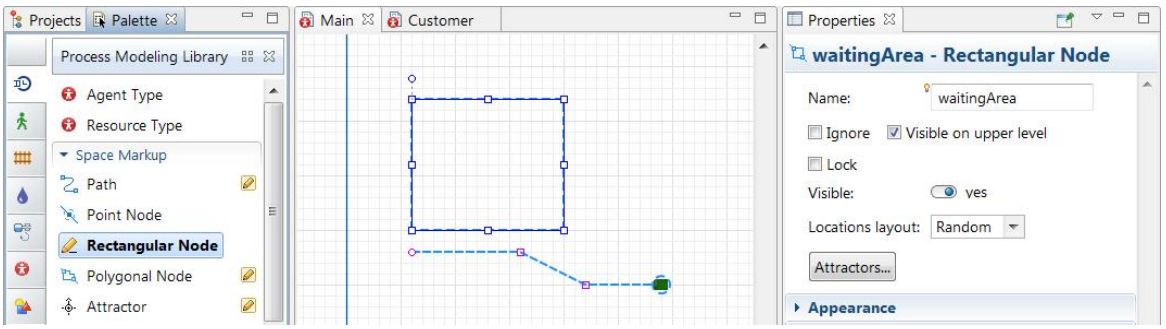
    Now we want to draw the area for queueing and a place to get serviced for our clients.

15. This time we will draw a waiting area using a rectangular node. First, open the **Space Markup** palette in the **Palette** view.

16. Double-click the element **Rectangular node** to switch to the *drawing mode*.

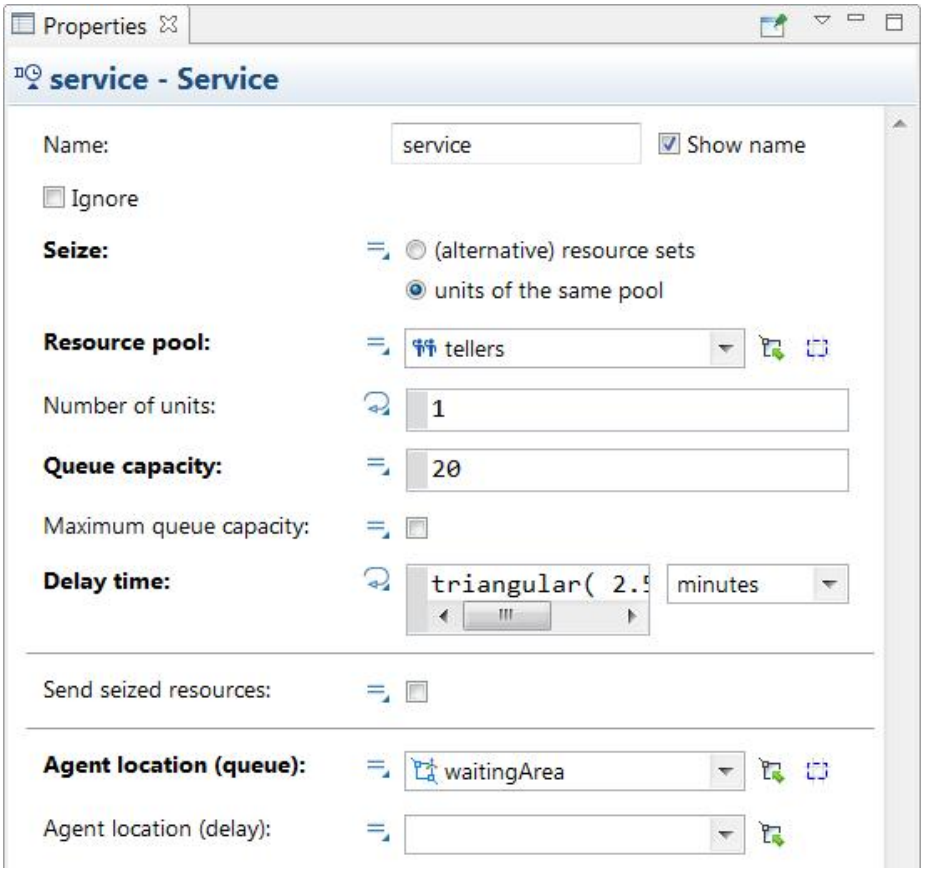
17. Click in the graphical editor and drag the rectangle without releasing the mouse button. Release when you have a rectangular node of the required form. You can edit its form later as you need.

18. Name the node *waitingArea*.



19. Click the *service* block in the flowchart and go to its **Properties** view.

20. Select the node *waitingArea* we have drawn in the **Agent location (queue)** option.

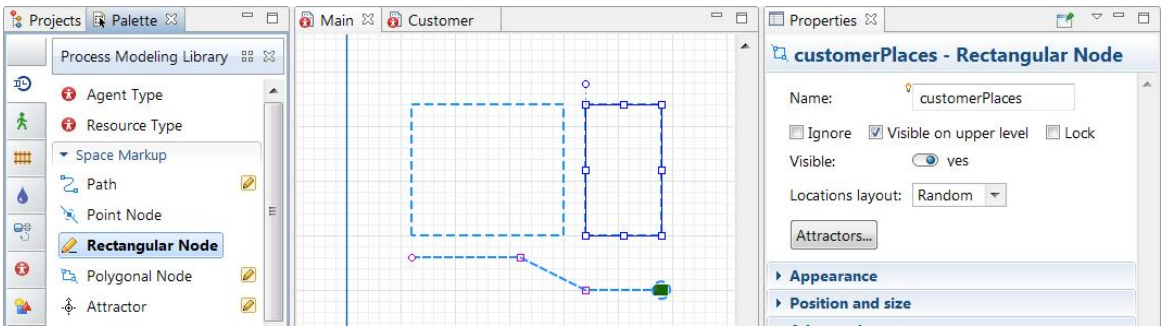


21. The customers need a place to stand somewhere while they are getting serviced by tellers. We will draw an area for this purpose using a rectangular node.

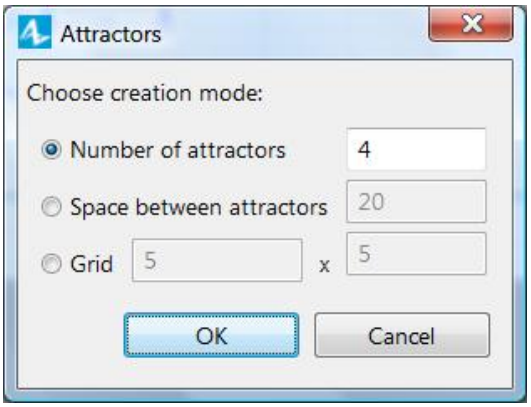
22. First, open the **Space Markup** palette in the **Palette** view, then double-click the element **Rectangular node** to switch to the *drawing mode*.

23. Click in the graphical editor and drag the rectangle without releasing the mouse button. Release when you have a rectangular node of the required form. You can edit its form later as you need.

24. Name the node *customerPlaces*.

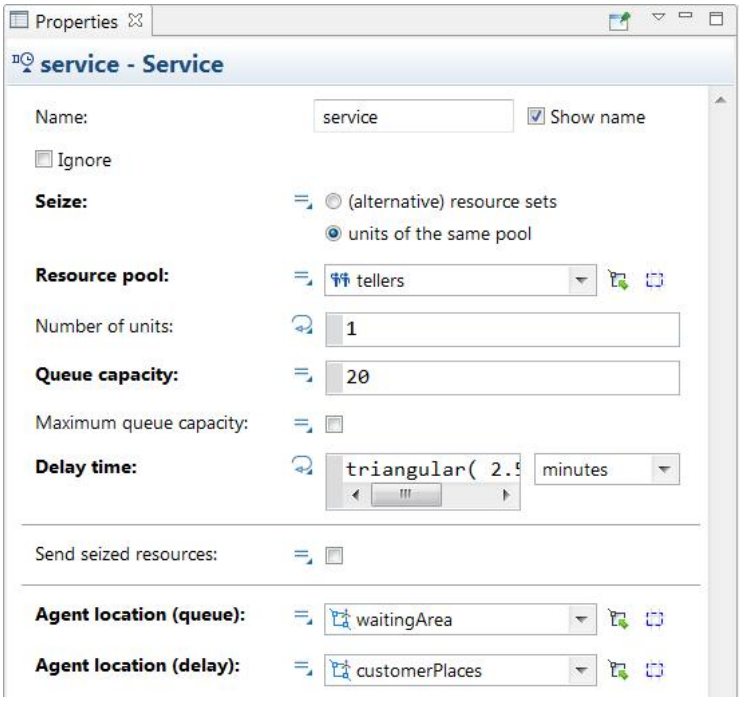


25. We will use attractors to define the customers that are getting service. Select the node *customerPlaces* and click the button **Attractors...** in its properties. In the **Attractors** window that will pop-up, specify *4* for the creation mode **Number of attractors** and click **OK**. You will see that attractors appeared in the *customerPlaces* node with the even offset.



26. Now we need to refer to this area in the flowchart. Click the *service* object in the flowchart and go to its **Properties** view.

27. Select the node *customerPlaces* we have drawn in the **Agent location (delay)** option.

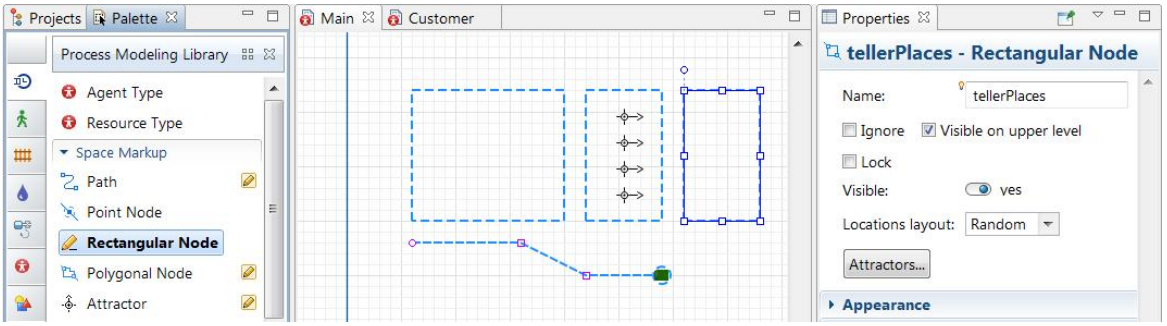


28. In the previous step, we used a point node to draw the ATM. Since we have 4 tellers this time, we will use a rectangular node to draw this service area.

29. First, open the **Space Markup** palette in the Palette view, then double-click the element **Rectangular node** to switch to *the drawing mode*.

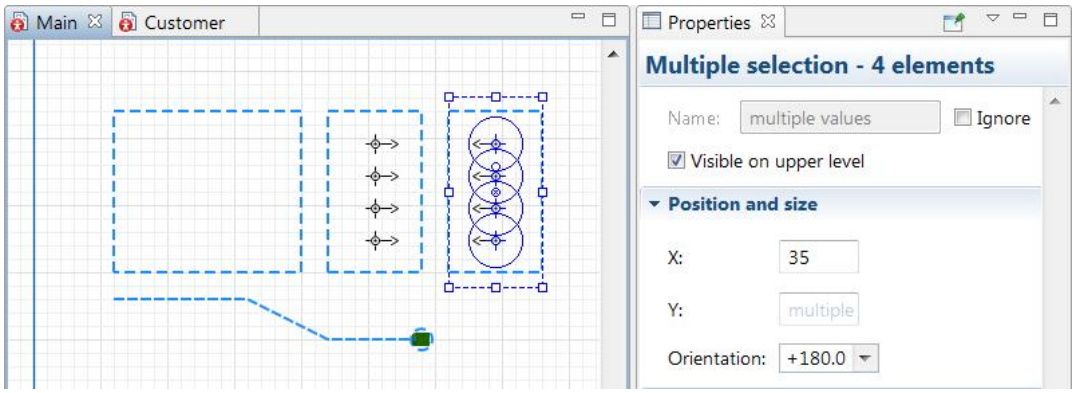
30. Click in the graphical editor and drag the rectangle without releasing the mouse button. Release when you have a rectangular node of the required form. You can edit its form later as you need.

31. Name the node *tellerPlaces*.



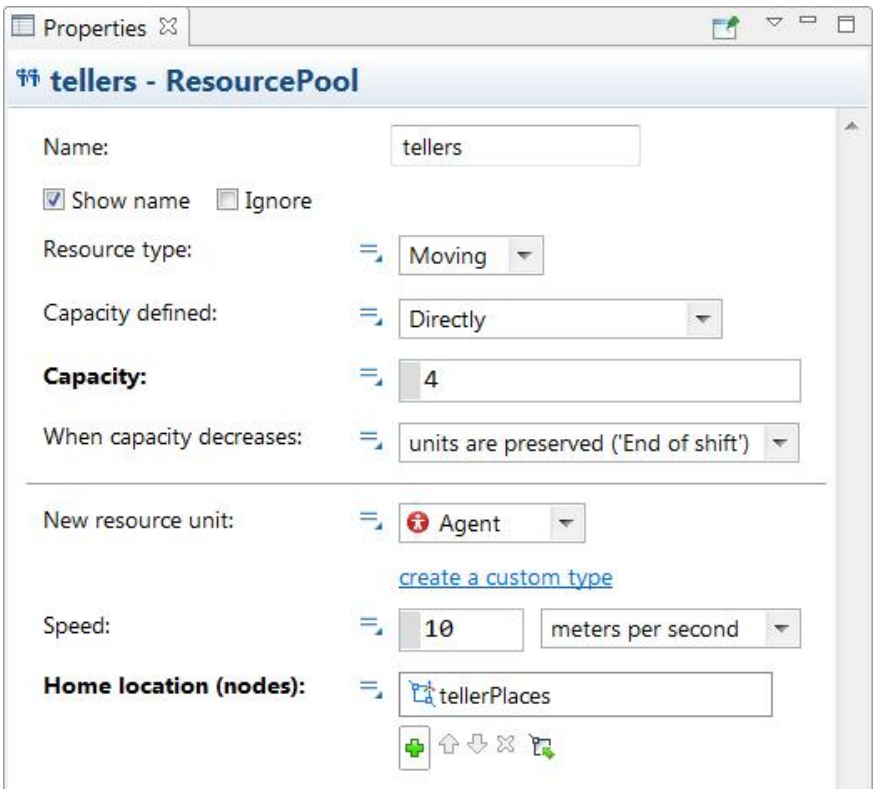
32. We will use attractors to define the tellers. Select *tellerPlaces* and click the button **Attractors...** in its properties. In the **Attractors** window that will pop-up, specify *4* for the creation mode **Number of attractors** and click **OK**.

33. You will see that attractors appeared in the *tellerPlaces* node with the even offset, but they are facing wrong direction. Select all attractors by Shift+clicking and go to their properties. In the section **Position and size**, change the **Orientation** parameter to *+180.0*.



34. Click the *tellers* object in the flowchart and go to its **Properties** view.

35. Select the node *tellerPlaces* we have drawn in the **Home location (nodes)** parameter.

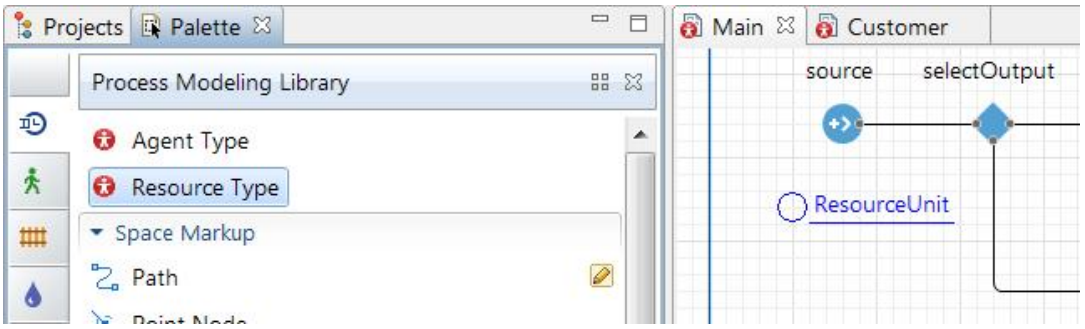


You can run the model now and observe how some customers are getting serviced at the ATM and some go to see tellers.

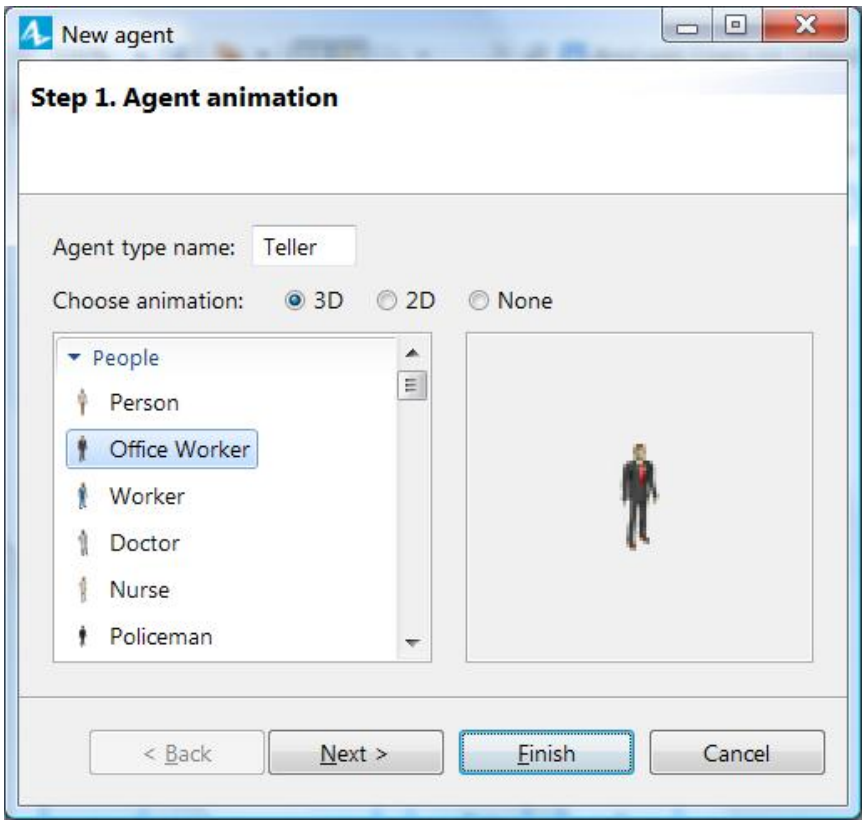
It is time to add teller 3D objects to our model. We will create a new resource type to animate tellers.

36. Open the **Process Modeling Library** in the **Palette** view.

37. Drag the element **Resource type** into the graphical editor.



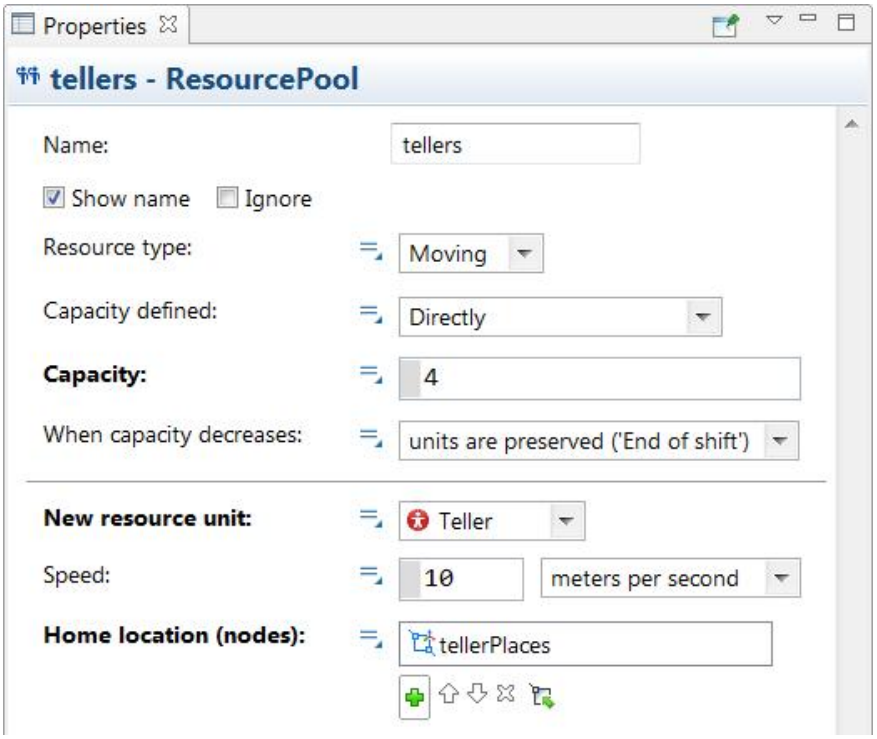
38. The **New agent** wizard will open on the **Agent animation** step. Enter *Teller* as the **Agent type name**, select **3D** as the animation type and select *Office worker* from the list of the 3D figures.



39. Click **Finish**. The new *Teller* diagram will open. You can find the *Office worker* 3D figure in the axis origin. Switch back to *Main* diagram.

40. On the *Main* diagram, select the block *tellers* in the graphical editor.

41. In its properties, specify *Teller* as the **New resource unit**.

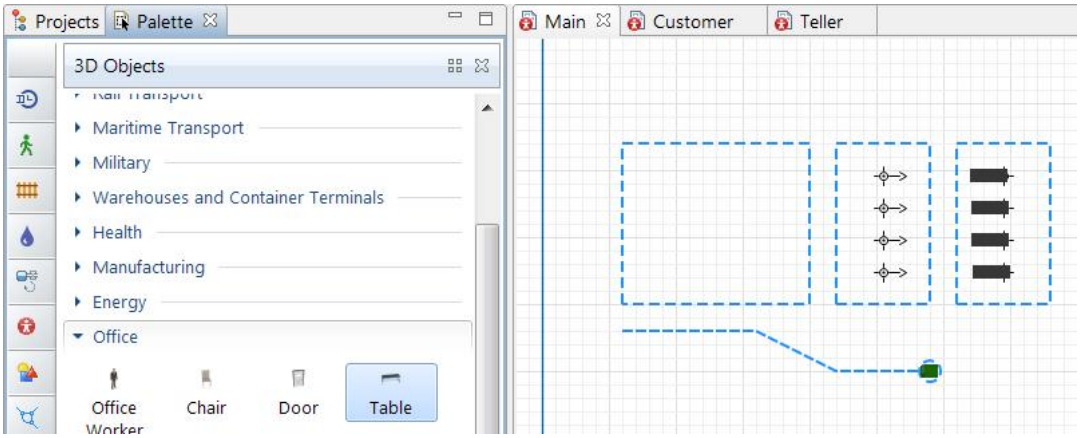


42. Run the model and observe customers and tellers.

43. Open the **3D Objects** palette in the **Palette** view.

44. Drag four **Table** 3D figures from the **Office** section of this palette onto the node shape called *tellerPlaces* in the graphical editor.

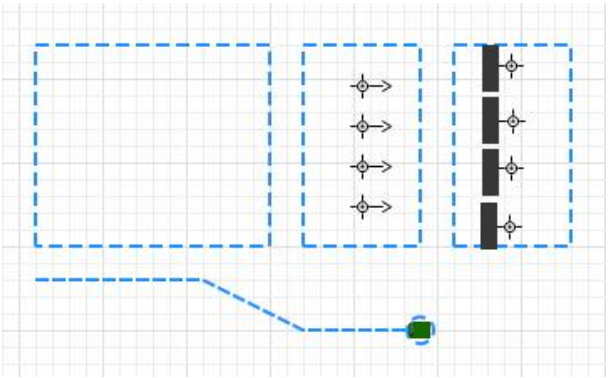
45. Place them at the attractors since attractors are the places where the tellers stand.



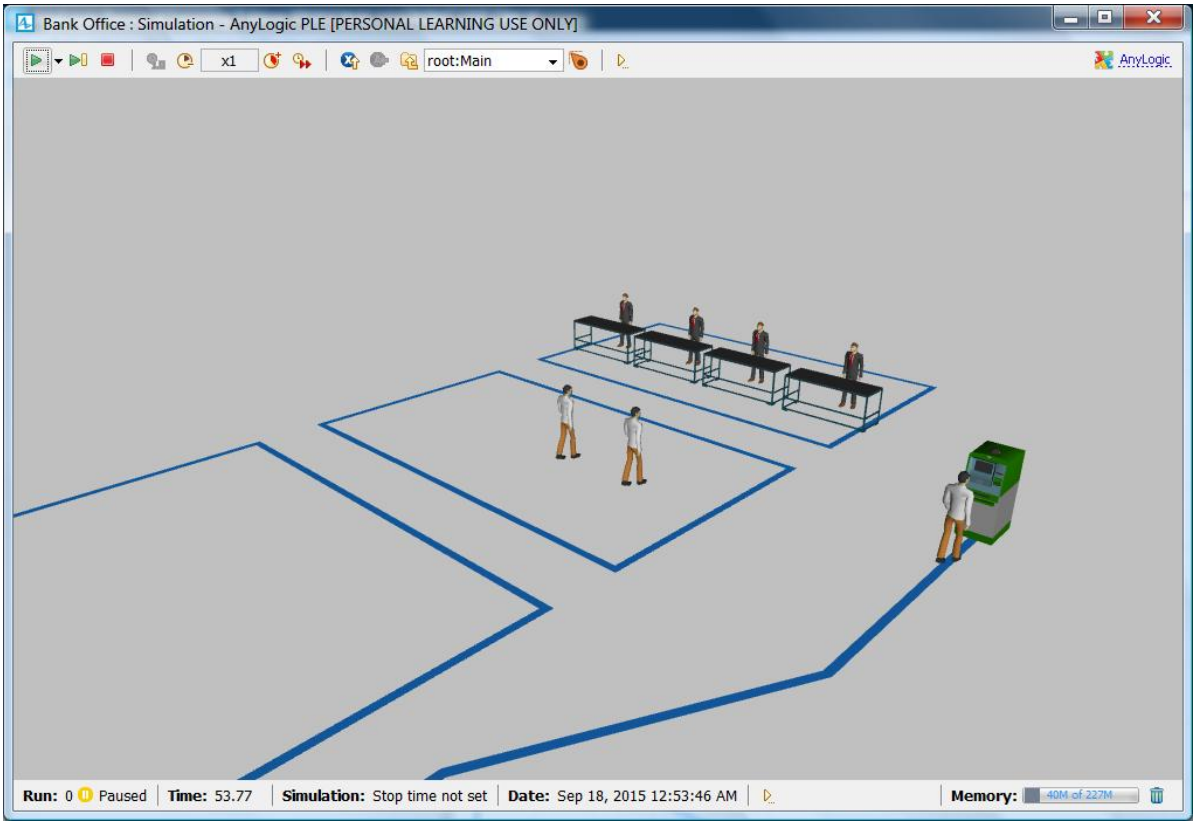
46. You can see that their orientation is wrong. Select all tables by Shift-clicking and go to the **Properties** view.

47. In the section **Position**, change the parameter **Rotation** to *-90.0* degrees.

48. If necessary, rearrange all eight attractors and four tables so that they are reasonably lined up.



Now you can run the model and observe in 3D how some customers go to the ATM and other get service at the tellers tables.

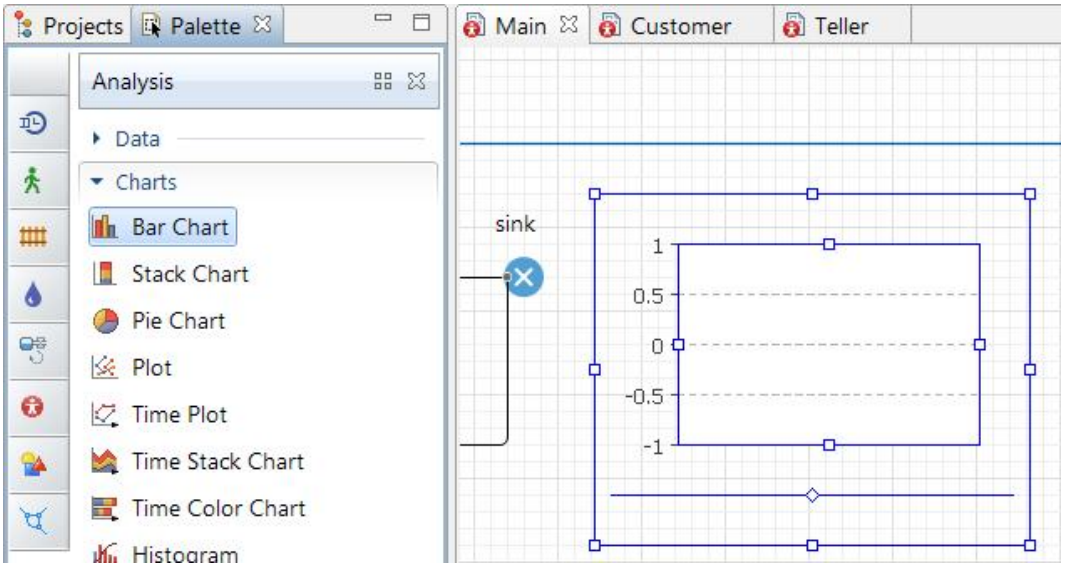


Phase 4. Adding Statistics Collection

With AnyLogic, you can collect complicated statistics whenever you need them. The objects of the **Process Modeling Library** are already capable of collecting the basic statistics. All you need is to turn the statistics collection for the object on, as it is disabled by default to speed up the model execution. We can view the statistics collected for the flowchart objects with charts.

We want to observe how mean ATM utilization and mean queue length change with time.

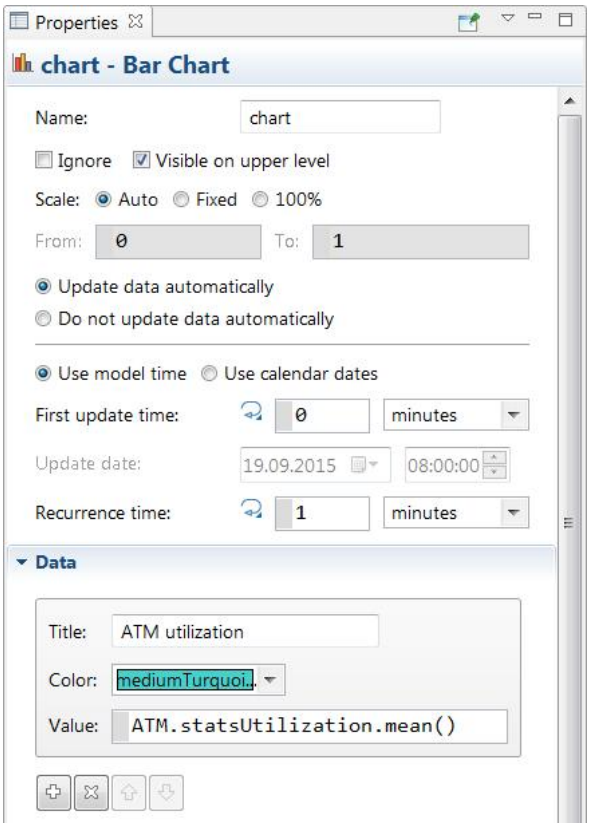
1. Open the **Analysis** palette of the **Palette** view. This palette contains charts and data objects used for collecting data and performing various statistical analysis on them. Drag the **Bar Chart** element from the stencil into the graphical editor.



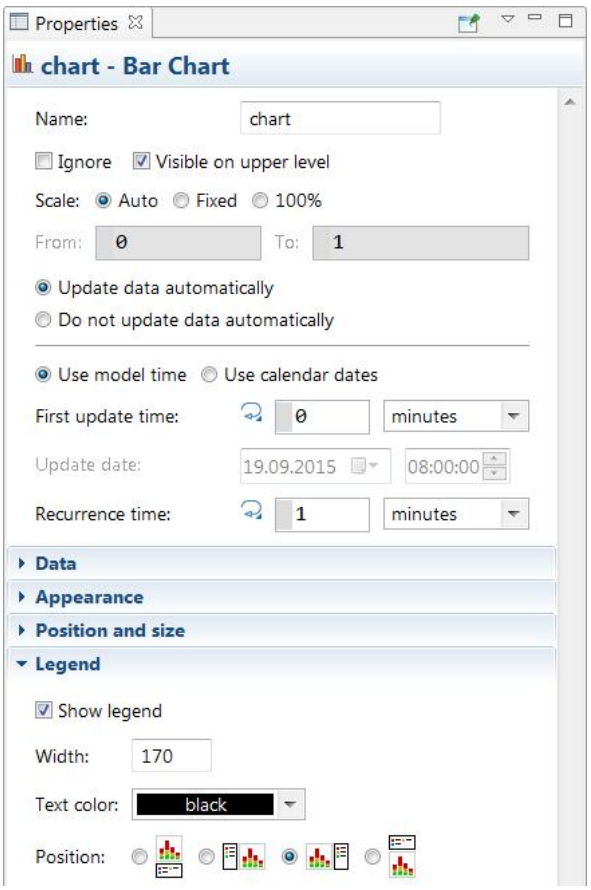
2. Go to the **Data** section in **Properties** of the chart. Click  **Add data item** to add data item to be displayed by this chart.

3. Modify the data item's **Title**: *ATM utilization*.

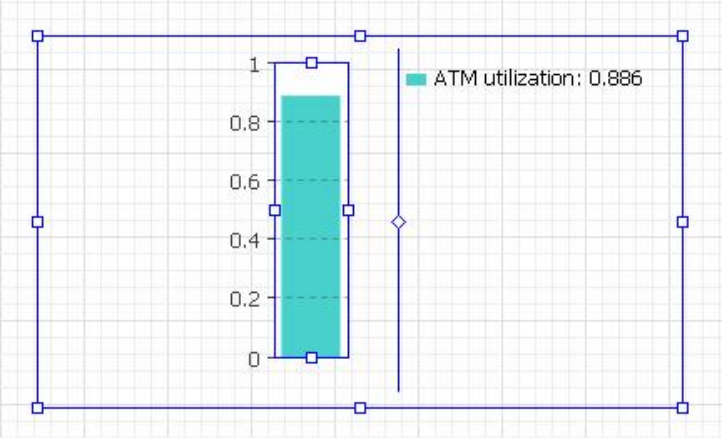
4. Type *ATM.statsUtilization.mean()* as the **Value** of the data item. *ATM* is the name of the **Delay** object we created. Each **Delay** object has *statsUtilization* data set that collects statistics on the object utilization. The *mean()* is the function that returns the mean value measured. You can use other functions to get statistical values, such as *min()* and *max()*.



5. Change the position of the chart's legend in the **Legend** properties section.

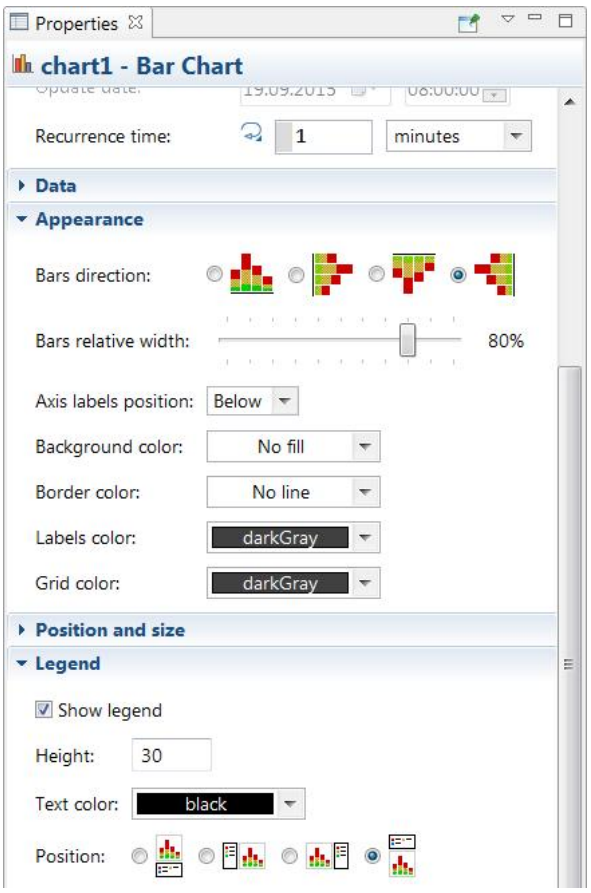


6. Then resize the chart as shown in the figure below:

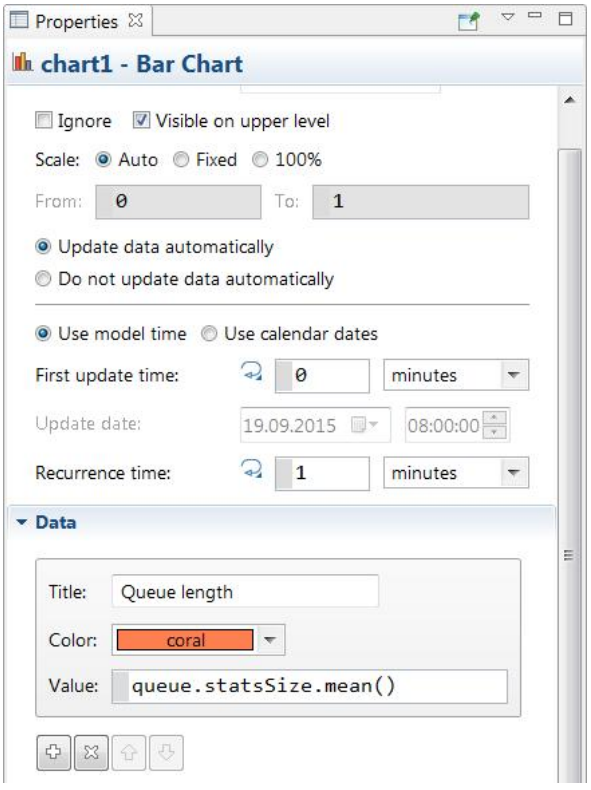


7. Add one more bar chart in the same way.

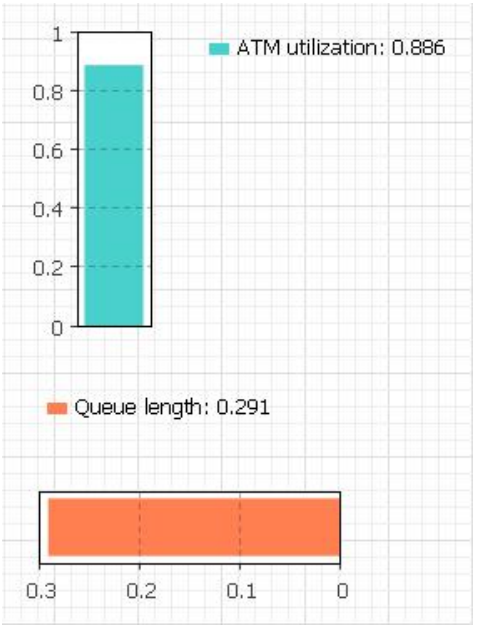
8. Open the **Appearance** section of the **Properties** view and choose the last option from the **Bars Direction** choice to make bars grow to the left and also change the position of the chart's legend in the section **Legend** (like it is shown in the figure below).



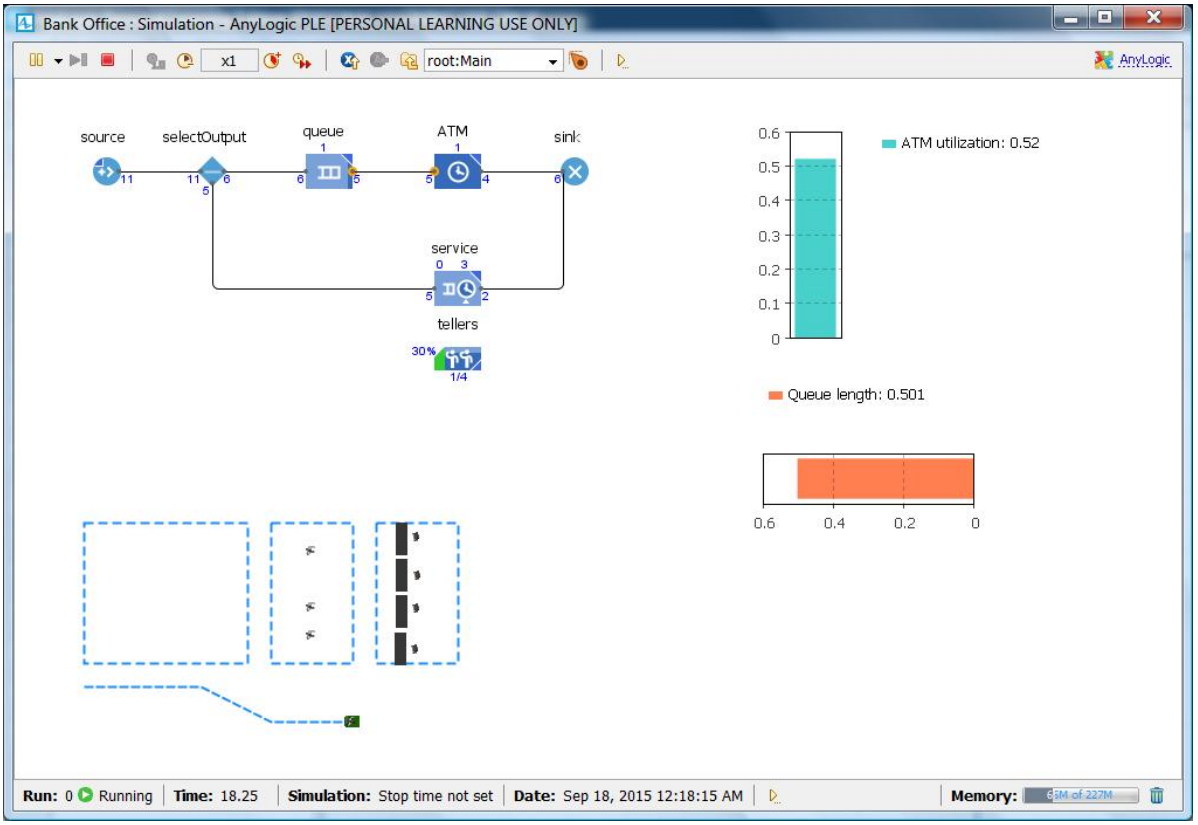
9. Add a data item to be displayed by the chart. Set **Title**: *Queue length* and **Value**: *queue.statsSize.mean()*. Here *statsSize* is the data set of type **StatisticsContinuous** that collects the statistics on the **Queue** size.



Resize the chart to look like the one in the figure.



Run the model and observe the ATM utilization and mean queue length with just created charts.



We want to know how much time customer spends waiting in ATM queue and the whole time he spends in the bank. We will collect time statistics using AnyLogic analysis data objects and observe the resulting time distributions using histograms. We will use the agent type *Customer* we have created on the Step 2.

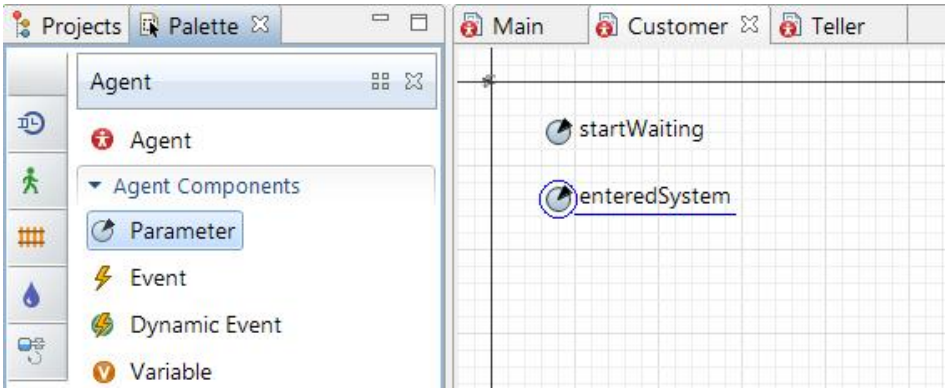
First we need to add two parameters into our model.

10. Switch to the **Projects** view. Double-click the agent type *Customer* to open its diagram. We need to create parameteres for the agent type *Customer* if we want to collect customer statistics.

11. Open the **Agent** palette of the **Palette** view.

12. Drag the element **Parameter** into the diagram of *Customer*.

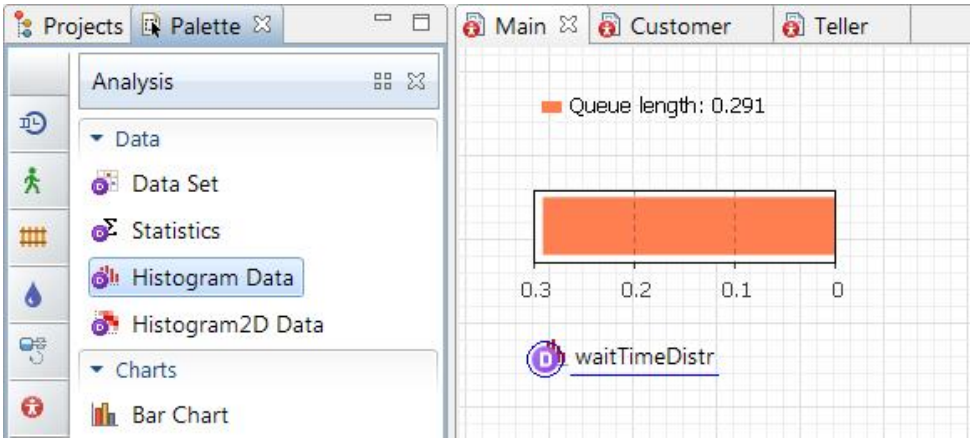
13. Leave the type *double* as it stands by default and name the parameters *startWaiting* and *enteredSystem*.



14. We will continue developing our model in the *Main* diagram.

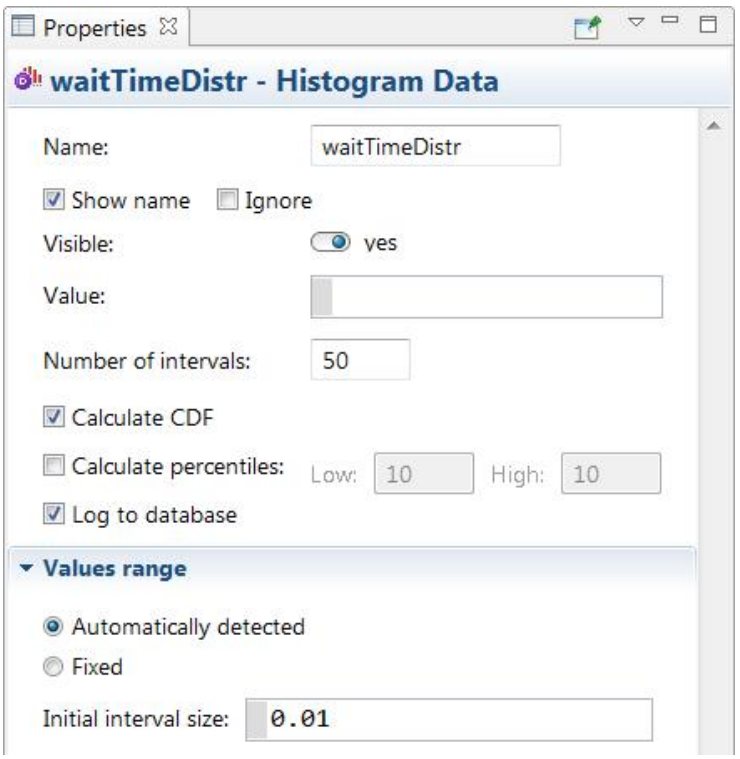
    Add histogram data objects to store statistics on customer's waiting time and time in system. Histogram data objects support standard statistical analysis on the data values being added (calculate mean, minimum, maximum, deviation, variance and mean confidence interval).

15. To add histogram data object on the diagram, drag the **Histogram Data** from the **Analysis** palette of the **Palette** onto the *Main* diagram.

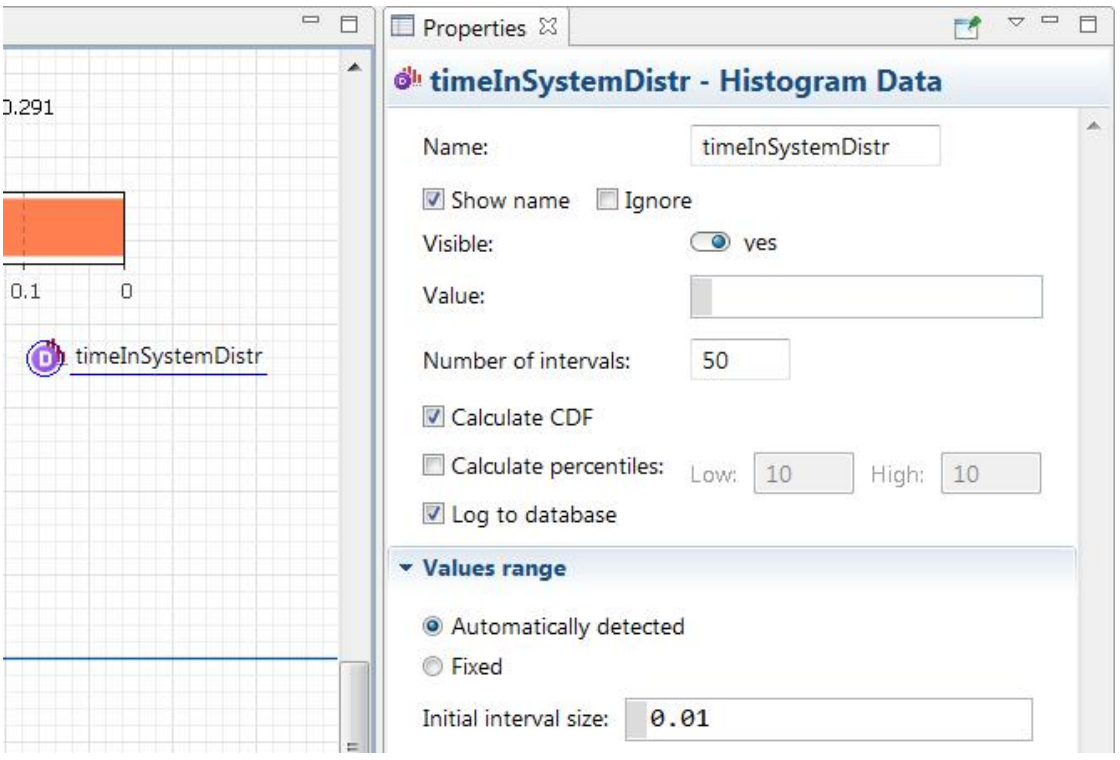


16. Set up the properties of the element.

    - Change the **Name** to *waitTimeDistr*.
    - Set the **Number of intervals** equal to *50*.
    - Set the **Initial interval size**: *0.01*.



17. Create one more histogram data object. Ctrl+drag histogram data object to create its copy. Change the **Name** to *timeInSystemDistr*.

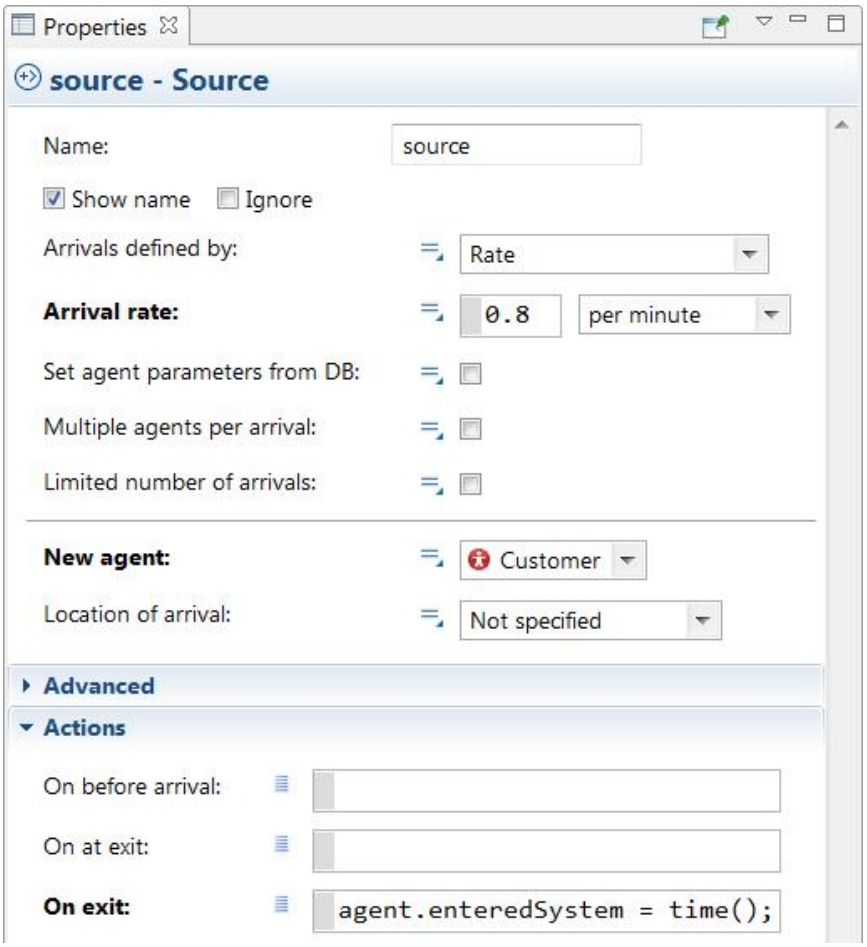


Now we will modify properties of our flowchart objects.

18. Modify source properties:

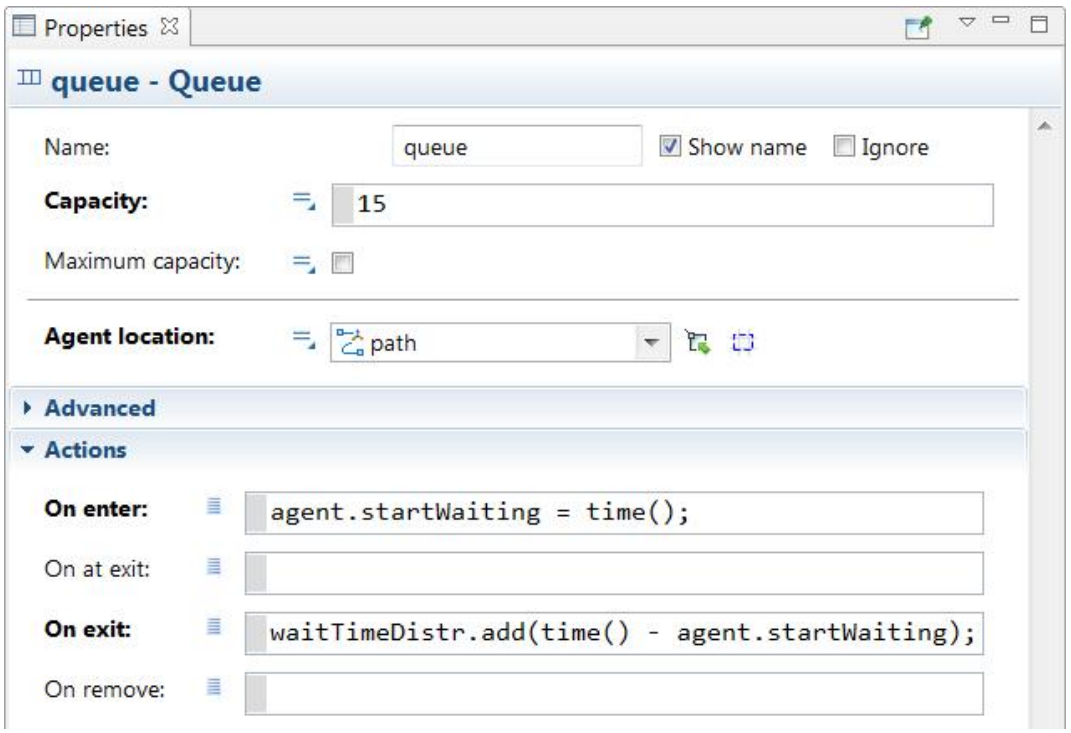
    - Make sure the agent type *Customer* is specified as **New agent**. This object should continue generating agents of our *Customer* type.

- Type *agent.enteredSystem = time();* in **On exit** action, located in **Actions** section. This code stores the time when a customer was generated in the *Customer's* variable *enteredSystem*. The *time()* function returns the current model time value.



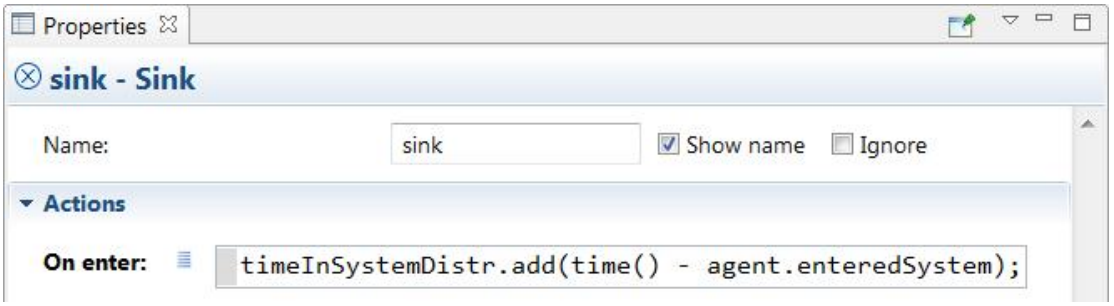
19. Modify queue properties:

- Type *agent.startWaiting = time();* in **On enter** action, located in **Actions** section. This code stores the time when a customer started waiting in the queue in the *Customer's* variable *startWaiting*.
- Type *waitTimeDistr.add(time() - agent.startWaiting);* in **On exit**. This code adds waiting time of the customer to the *waitTimeDistr* histogram data object.

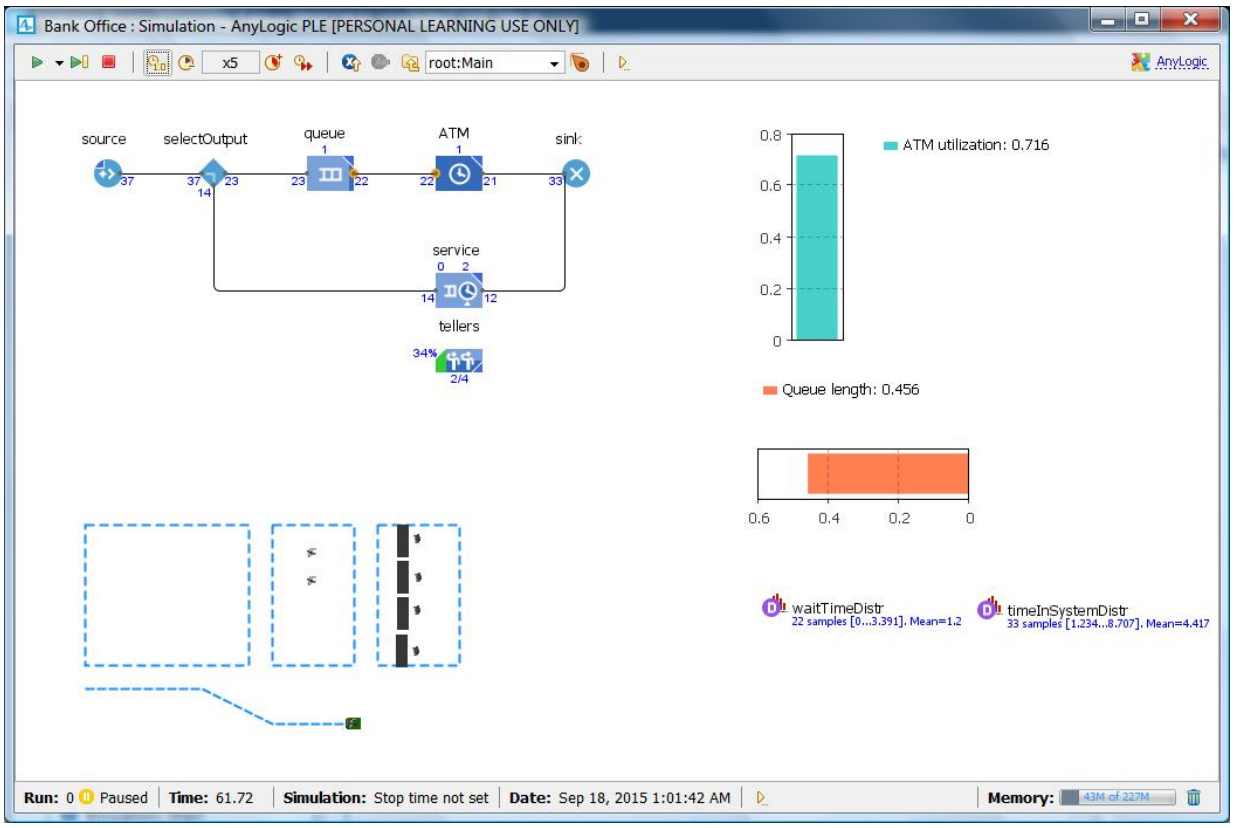


20. Modify *sink* properties:

- Type *timeInSystemDistr.add(time() - agent.enteredSystem);* in **On enter** action, located in **Actions** section. This code adds the whole time the customer spent in the bank to the *timeInSystemDistr* histogram data object.

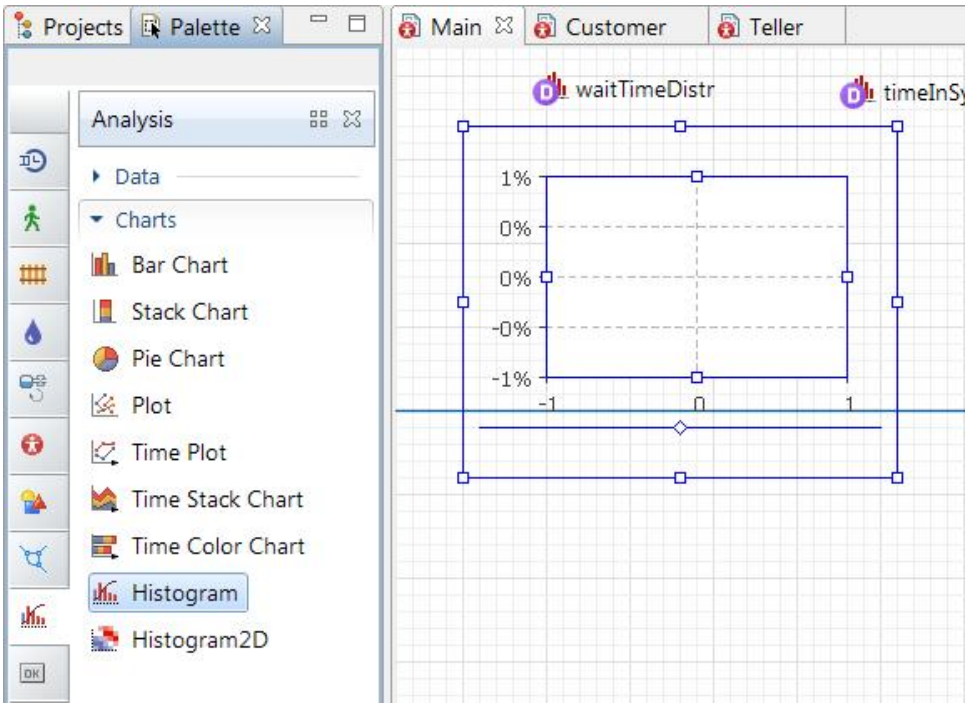


Run the model and view the statistics using inspect window of the data set. Open inspect window for data set by clicking on it. Here you can see standard statistical analysis on the data values being added to this data object.

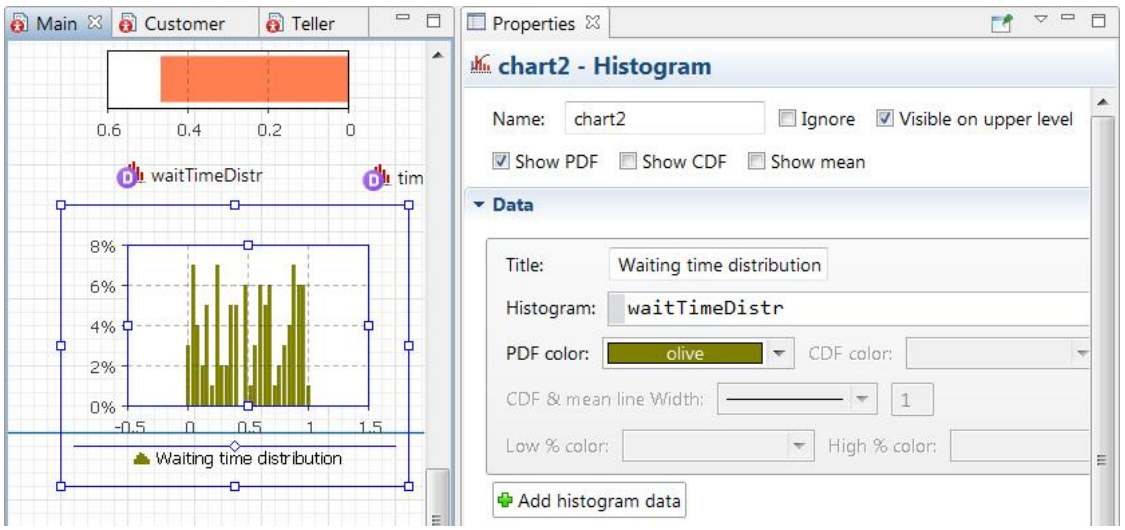


Now we want to display the collected statistics using standard histograms.

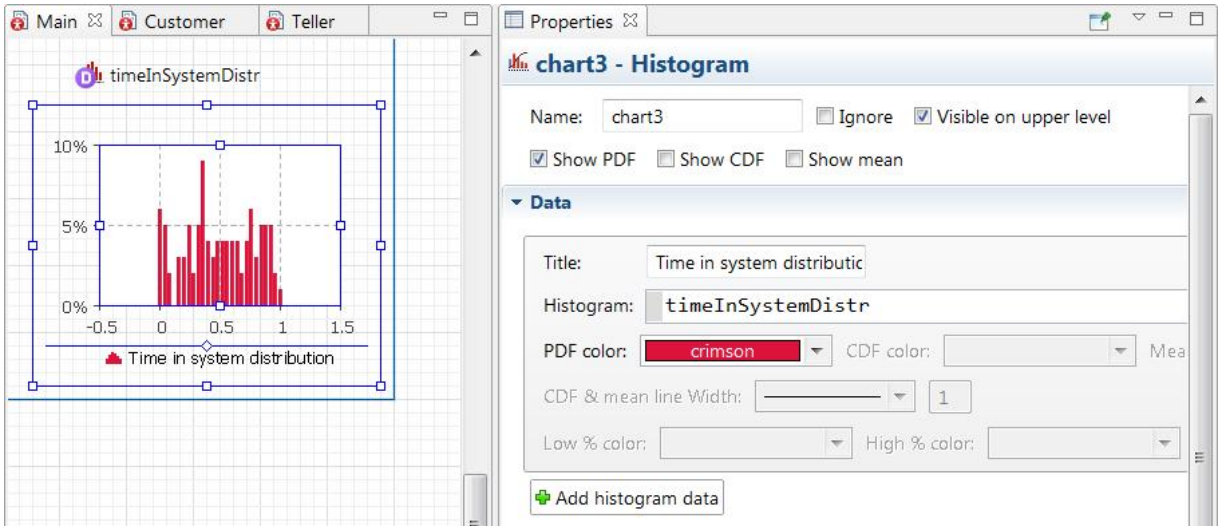
21. To add histogram on the diagram, drag the **Histogram** element from the **Analysis** palette of the **Palette** onto the diagram where you want to place the histogram. Resize it if needed.



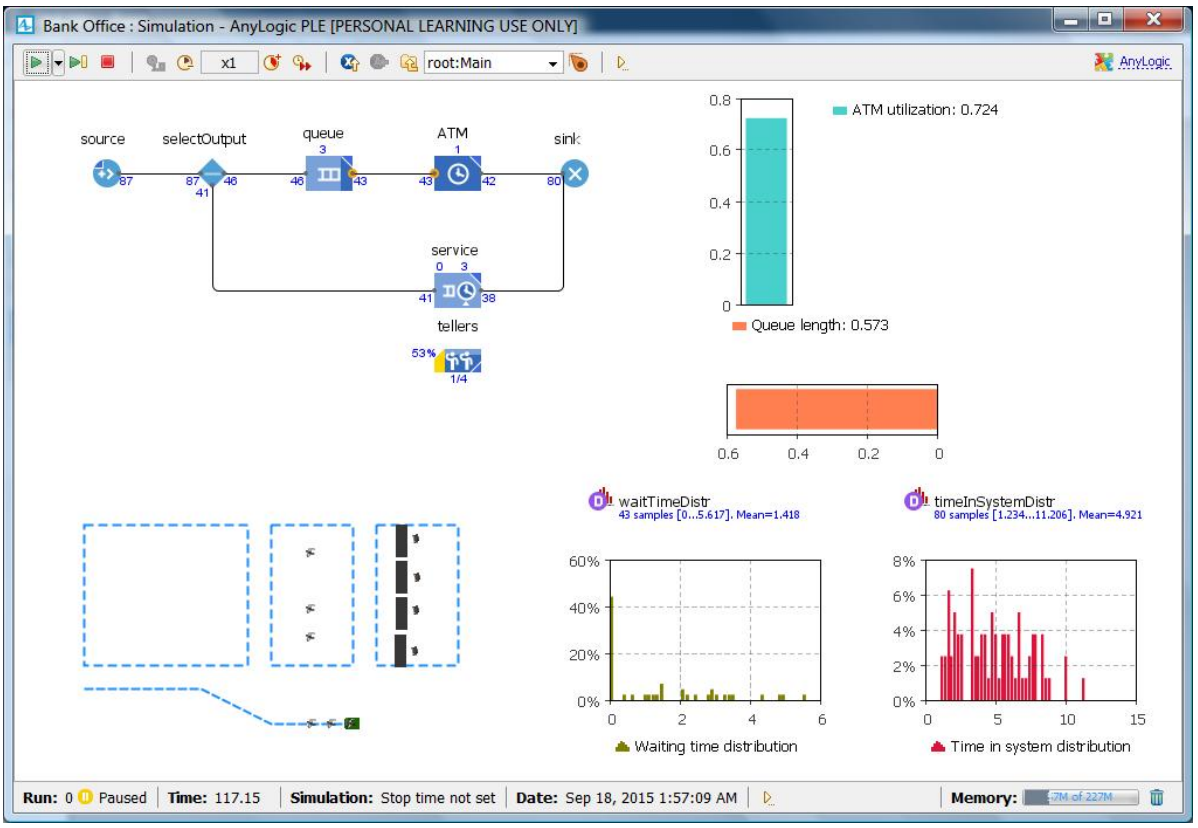
22. Define the data object to be displayed by this histogram. Click the **Add histogram data** button in **Data** section and change the **Title** to *Waiting time distribution*. Specify the **Histogram** to be displayed: *waitTimeDistr*.



23. Add one more histogram below the existing one. Change the **Title** of the displayed data to *Time in system distribution.* Choose the data object to be displayed: *timeInSystemDistr.*



24. Run the model. Set virtual time mode and observe distribution of customer's time in system and waiting time.


Bank Office : Simulation - AnyLogic PLE [PERSONAL LEARNING USE ONLY]
x1
root:Main
AnyLogic
source
selectOutput
queue
ATM
sink
service
tellers
ATM utilization: 0.724
Queue length: 0.573
waitTimeDistr
43 samples [0...5.617], Mean=1.418
timeInSystemDistr
80 samples [1.234...11.206], Mean=4.921
Waiting time distribution
Time in system distribution
Run: 0 Paused
Time: 117.15
Simulation: Stop time not set
Date: Sep 18, 2015 1:57:09 AM
Memory: